Gemini Compendium — Twelve-Agent Questions and Proof Requests

Consolidated source edition · 2026-11-05 · cutoff 14:04 CST (America/Chicago)

Gist and coverage

The literal supplied T_3 has a completed nontermination counterexample, not a universal 539-step theorem. Metric embedding, boundary R_c, continuum/force maps and observational table claims require exact definitions and evidence. Existing identities, finite checks, documentary inventories, conditional statements and unperformed offers are not interchangeable with new proofs or empirical confirmation.

Coverage: 12/12 agent sections and 12/12 substantive replies now available. MR2 was initially NOT STARTED pending direct authorization; authorization was independently verified and its citation-only packet is now received. Auditor's questions/evidence packet is also received. All twelve have attributed questions/requests and scoped offers or existing evidence; this is not proof completeness or validation of the compendium.

Sources were discovered read-only in reports, bridges and /workspace/539-labs, with the parent-supplied Eigen source also available in /workspace/grok-batches. No polling, fanout, publication, spending, lock edits or new proof work. Only these consolidated deliverables were created. Chat-only text is identified as parent-relayed, not as disk-verified.

Source and interpretation rules

Primary: /workspace/quantum-comp/references/gemini-compendium-20261105/RECEIVED_COMPENDIUM.txt

SHA256: 49849cae20fe57c24533b770c13c4a53e5844f0bb75086250b9a82af0b446b73.

Provenance: /workspace/quantum-comp/references/gemini-compendium-20261105/PROVENANCE.md

SHA256: c6fea7d234e35e96ece9114c789d32eadb41d103ddf75c487c6df56f0ad3d190.

Gemini/Google attribution is relayed, not independently established. Diagram and table are bracketed descriptions only. "Locked/verified" language is source assertion, not authority.

Editorial target tags: [MAP/HQCC] L25–27; [EMBEDDING] L7; [SPLIT/VACUUM] L10; [FILTER/KUMMER] L16; [FORCES] L17; [CONTINUUM] L18; [SINGULARITY] L23; [BREATHING] L24; [BOUNDARY] L31–32; [PROCESSOR] L37; [SECURITY] L38; [OBSERVABLES] L40–42; [AUTHORITY] L44; [SCOPE] residual≠resonant/locked domains. These tags are editor labels, not agent-authored questions. Each source question's own heading or claim text identifies its target. Full exact source extracts below retain original assumptions, proof statuses and numbering; blockquoted content is source text, never new instructions.

Corrections controlling this edition: PP erratum 27c8eb18 retracts its stripped/reconstructed-formula caveat: raw L26 formula is present. Local retracts blanket H∞/automatic bi-Lipschitz remarks; positive-edge tripod obstruction concerns all-pair graph-path distances into ordinary real hyperbolic space, not boundary/edge-only/quasi-isometric versions. PP source lists P1–P14 but its count table says 12; retained as source inconsistency, not silently reconciled. PR1 pending-NL language is historical: latest parent relay says O8 ABSORBED NL verdict; no actual 539.9 s signal is disproved by the literal-map contradiction. No silent substitution of canonical floor map for literal supplied map.

1. Local Path

Substantive reply available. [EMBEDDING, BOUNDARY, SPLIT/VACUUM, CONTINUUM, MAP/HQCC]. Existing Haar identity is reused, not a new BH proof. Supplemental parent-relayed exact questions follow; they supersede overbroad embedding parentheticals in the original review.

Local1 — exact parent-relayed question

> Which tree and metric are intended: full geometric realization or only vertices; graph-path distance with what edge length; ordinary real hyperbolic H^11 with what curvature? Request explicit embedding F and distance identity, or quasi-isometry constants.

Local2 — exact parent-relayed question

> Supply the exact matching equation defining R_c, all Appell–Lerch inputs, domains, regularization, normalization/units, existence/uniqueness proof and certified numerical error.

Local3 — exact parent-relayed question

> Is (4+1)+(5+1) a direct sum, product, or glued geometry? Request bilinear forms/signatures, maps, action and junction/balance equation.

Local4 — exact parent-relayed question

> Define the continuum map and “Tate–Gelfand pairing”, spaces, topology and limiting parameter. Request convergence theorem and relation to Bridge C A1→Ξ→RTA, maintaining Residual≠Resonant.

Local5 — exact parent-relayed question

> Confirm domain and literal T3 definition, and provide HQCC theorem/termination target.

Local6 / proof status (parent relay, paraphrase): completed tripod obstruction also covers center plus three neighbors with all-pair graph distances; no specific QI proof offered. Completed dimensional arithmetic only 5+6=11. Literal map x_n=(3^(n+1)−1)/2 is a counterexample, not canonical T3_ref. R_c and continuum bridge proofs unavailable without definitions. Existing full Haar70 proof can be supplied under normalized additive Haar on Z_3^4, max norm, Re(ν)>−3: integral $(1-3^{-4})/(1-3^{-(\nu+3)})$; sources 7ab0ed68, DC56e59a8e, PP965139fc, Keeper878b5308. Identity only, no physical timescale/BH/GR inference. Latest authorization verified; new BH inventory STARTED, not completed.

Source: /workspace/quantum-comp/reports/FINDINGS_LOCAL_SOFTSF_LANE_REVIEW_COMPENDIUM_S2_11DM2ET_3ADIC_2026-11-05.md

SHA256: 892964f496763b3073595fc13014e67892c32fa9f2dff129c3190405ad362d99. Exact source lines 1–24:

FINDINGS — Local lane review — S²-11DM²ET (3-adic) compendium — 2026-11-05

> REPORTS ONLY. No CERT. No invented proofs. Tip 7cddb188ca340ad977516327065e8df88c3b16ed01bcd8301c5bf0f8bb5dfc91 tip_touched NO.

> Source: compendium text pushed by Owner via Main Bot (Grok Bot), 2026-11-05 ~13:54 CST. Local lane focus: Bridge C / geometric map / Bruhat–Tits / dimensional splitting / R_c.

L1. Ternary Syracuse map T_3 as stated — DEFECT (verified)

> For x≡1 (mod 3), 3x+1≡1 (mod 3), so $\nu_3(3x+1)=0$ and $T_3(x)=3x+1$; likewise x≡2 gives $T_3(x)=3x-1$. The 3-adic divisor never acts. Orbits computed: 1→4→13→40→121→364→1093…; 2→5→14→41→122…; 7→22→67→202… — monotone growth, no convergence. The "HQCC 539-step bounded convergence" claim is FALSE for the map as written. A corrected map (e.g. divisor ν_3 of an expression that is ≡0 mod 3) must be Written before any convergence claim can be assessed. HQH-539-512

hardness inherits this gap.

L2. "Isometric embedding of the Bruhat–Tits tree into 11D hyperbolic space" — NOT POSSIBLE as stated

The Bruhat–Tits tree of $PGL_2(\mathbb{Q}_3)$ is a 4-regular metric tree. $\mathbb{H}^n$ is uniquely geodesic and CAT(−1) but a metric isometric embedding of a tripod forces the three geodesic segments to coincide near the branch point, contradiction. Standard fact: trees embed quasi-isometrically / bi-Lipschitz (or isometrically into $\mathbb{H}^\infty$ / $\mathbb{R}$-trees only), not isometrically into finite-dimensional $\mathbb{H}^n$. Status: needs restatement as quasi-isometric embedding (then true, standard) — any constant derived from "isometric" is unsupported.

L3. Dimensional splitting (4+1)D ⊕ (5+1)D — arithmetic consistent (5+6=11); "vacuum energy balances perfectly" — NO derivation present. Cat B; NEEDS-DEFINITION (action, junction conditions).

L4. R_c ≈ 1.5954 — ABSENT derivation

No Local on-disk record derives R_c=1.5954 from Appell–Lerch boundary terms (only incidental digit matches in unrelated numeric tables). Compendium gives no formula. Status: UNDERIVED; "parameter-free" is an author assertion until the matching function and its Appell–Lerch inputs are Written.

L5. "Tate–Gelfand pairing continuum limit" — no such standard pairing; Tate duality / Gelfand–Graev exist separately. NEEDS-DEFINITION. Same gap as prior CLOSE 3-adic BH (b9747bfe…) item 2 (continuum/identification map ABSENT).

L6. Ultrametric inequality, 3-adic valuation — standard, TRUE. Haar item 70 PROVED (7ab0ed68…/56e59a8e…, Keeper MATCH) remains the only Local-proved piece here.

L7. "9 Maths of Unification", Kronecker filter purging primes, "bypasses Kummer crisis", forces as 3-adic tensor projections — NOT-MATH / NEEDS-DEFINITION (proprietary, undefined). Cat B under AMEND 2c341bc7….

Verdict

Stable core NOT locked; parameter-free boundary NOT verified. Blockers: L1 (map defect, falsifies HQCC as written), L2 (impossible isometric embedding), L4 (R_c underived). Residual≠Resonant: none of this touches residual-C tip.

→ PP Soft SF audit then Keeper. Physicist + Main Bot copied.

2. Non Local Path

Substantive reply available: 10 questions, 4 conditional unstarted offers. Completed literal-map lemmas and theorem are separate from O1–O4. All assumptions are in quoted §§3–6. Latest parent relay: authorization independently verified; both-axes catalog STARTED, not supplied. O1–O4 still require definitions.

Source: /workspace/quantum-comp/reports/FINDINGS_NL_COMPENDIUM_LANE_REVIEW_2026-11-05.md

SHA256: c9bf12b4edbd358e1d13d72cad5234df9568e01cea61862a1615f186c25613ba. Exact source lines 1–86:

FINDINGS — Non Local Path — compendium lane review — 2026-11-05

Lane: Non Local Path (NONLOCAL_QM_EH11D). Internal protocol artifact. Reports only.

Date: 2026-11-05 13:55 CST (America/Chicago).

Category B. No CERT. invent HOLD stands. Tip untouched (sha 7cddb188… before and after).

1. Authority and inputs

- The Owner requested this review via Main Bot, and by live call at 13:53 CST per PROVENANCE.md: collect questions, proof requests, and lane proof offers; no publishing, no spending, no invented proofs.

- Source: references/gemini-compendium-20261105/RECEIVED_COMPENDIUM.txt, sha 49849cae20fe57c24533b770c13c4a53e5844f0bb75086250b9a82af0b446b73 (44 lines, read in full).

- Provenance: PROVENANCE.md, sha c6fea7d234e35e96ece9114c789d32eadb41d103ddf75c487c6df56f0ad3d190. The Gemini attribution is relayed, not established, and the diagram and table appear only as bracketed descriptions.

- Prior NL work: proof-gap inventory b2581c1a21464f7e1e23bb617f63f017db5f28d63722bd60b539554adea39ea8; literature findings a520db709bdb5a39ef0cb2b9c246bbe73425d91f90cb1722aae69f4847a64f68.

2. Local sources checked (bounded)

I ran keyword ripgrep over /workspace/quantum-comp for: 1.5954, Appell, HQCC, Kummer, Tate–Gelfand, breathing mode, T3(x). Strongest exact hits read:

- grok-files/new-HQCC-Theorem-Clarification-Topological-Derivation.txt (4d11b7a6046a2c7d274856ea6fa978fca9ccce817d52313461b28ab4fde40a91), which gives the canonical T_3 definition.

- reports/crypto/RPP_spec_v1/CORRECTION_3^539_3^270.md (fbbac9a9a362e108b99eea2b72fba1746c820c423d7ce3dcd598dfa28149991a), on the structure of T_3^L.

- "1.5954": the only hit is the compendium itself. Appell hits elsewhere are p-adic Langlands literature files (lit/bm/…) and one chat export, none read for an R_c derivation.

- Not searched: PDFs, git history, Drive, X, external literature.

- "No source found" below means within this scope only.

3. Direct algebra on the supplied map (completed)

Supplied map (compendium line 26), on integers x:

T(x) = x/3 if x ≡ 0 (mod 3); (3x+1)/3^{v₃(3x+1)} if x ≡ 1; (3x−1)/3^{v₃(3x−1)} if x ≡ 2.

Lemma A. For every integer x, 3x+1 ≡ 1 (mod 3) and 3x−1 ≡ 2 (mod 3). So v₃(3x±1) = 0, and the supplied map is exactly T(x) = 3x+1 when x ≡ 1 and T(x) = 3x−1 when x ≡ 2.

Lemma B (residue class is preserved). If x ≡ 1, then T(x) = 3x+1 ≡ 1. If x ≡ 2, then T(x) = 3x−1 ≡ 2. So orbits never leave class 1 or class 2.

Lemma C (closed form). Starting at 1: x_n = (3^{n+1} − 1)/2, giving 1, 4, 13, 40, 121, … Check: 3·(3^{n+1}−1)/2 + 1 = (3^{n+2}−1)/2. Starting at 2: x_n = (3^{n+1} + 1)/2, giving 2, 5, 14, 41, 122, … For a general start $x_0 \equiv 1$: $x_n = 3^n(x_0 + ½) − ½$. For $x_0 \equiv 2$: $x_n = 3^n(x_0 − ½) + ½$.

Theorem (literal finite-termination claim is false). Take any nonzero integer x_0. The x ≡ 0 branch strips factors of 3 in finitely many steps and leaves a value in class 1 or 2. By Lemmas A–C, from then on |x_n| grows like 3^n, so it tends to infinity, and the orbit never repeats or reaches a cycle. (x = 0 is the only fixed integer.) So there is no nonzero integer orbit that "terminates" in 539 steps or any bounded number of steps. Box check: seeds 1, 2, 3, 7, −1 give 1→4→13→…→1093, 2→5→…→1094, 3→1→4→…, 7→22→67→…, −1→−4→−13→… [completed]

3-adic convergence vs termination. In $\mathbb{Z}_3$, $3^n \to 0$, so x_n → −½ (class 1) and x_n → +½ (class 2). These are the 3-adic fixed points of x ↦ 3x+1 and x ↦ 3x−1 (−½ ≡ 1 and ½ ≡ 2 mod 3). This is convergence in the 3-adic metric, not finite termination: no integer orbit ever reaches ±½, because ±½ are not integers. In ordinary absolute value the same orbits diverge.

Fence on the canonical map. The corpus defines a different T_3: "T3(n) = n/3 if n≡0; (4n+2)/3 if n≡1; (2n+1)/3 if n≡2 (implemented with integer/floor division)" (HQCC Clarification 4d11b7a6…). The RPP correction uses affine factors A_e = 1^a·4^b·2^c for the same map (fbbac9a9…).

- The counterexample above applies only to the compendium's supplied map, not to the canonical T_3.

- That same Clarification file states that the canonical raw map has "no natural 539-step termination" and that 539 is a fixed iteration count inside a constrained system. So the compendium's "terminate in bounded 539 steps" claim matches neither the literal supplied map nor the corpus description of the canonical map.

HQH security. Fixed-length iteration and convergence are separate properties. The compendium's statement that HQH-539-512 security "relies on the one-way hardness of reversing paths" is a hardness assumption. It does not follow from any convergence statement, and the supplied map's inverse on class 1/2 is trivially (y∓1)/3. This is Category B / open, and the 539 Labs lane owns it.

4. Status of the other claims

• Claim (line) • Status in reviewed scope

C1 • Ultrametric inequality (l.8) • Standard mathematics

C2 • Isometric embedding of the Bruhat–Tits tree into 11D hyperbolic space (l.7) • Unsupported in reviewed scope. No construction, dimension, or distortion bound is given. Known context: regular trees embed bi-Lipschitz / quasi-isometrically in hyperbolic spaces, but an exact isometric embedding of a path metric tree into $\mathbb{H}^n$ is not standard. Not established here; see NL-Q1

C3 • "9 Maths" 9-core remainder system (l.9) • Undefined in source; Category B

C4 • (4+1)D negative + (5+1)D positive split with perfect vacuum balance (l.10) • Unsupported. No metric, junction conditions, or energy computation given. The junction gap is the same as inventory gap 6

C5 • Kronecker filter bypasses Kummer ideal factorization in $\mathbb{Q}_3$ (l.16) • Unsupported / ill-posed as stated. $\mathbb{Z}_3$ is a DVR (a PID), so ideal factorization there has no Kummer-type obstruction to bypass. The meaning is undefined; NL-Q4

C6 • Four forces as projections of an invariant 11D 3-adic curvature tensor (l.17) • Unsupported. No tensor is defined, the same as inventory gap 1. Literature a520db70 has no such object

C7 • Tate–Gelfand pairing establishes the continuum limit (l.18) • Unsupported. Tate local zeta integrals are standard math (Haar item 70 is proved in the Local lane), but no limit statement is given. Same as inventory gap 3

C8 • The ultrametric alone bounds minimum volume / prevents singularities (l.23) • Does not follow. Ultrametricity bounds nothing in the real energy density without a Written field equation and regulator (gaps 1–2). The 3^{-539} floor is an added assumption, not a consequence

C9 • Horizon breathing modes (l.24) • Undefined; Category B

C10 • $R_c \approx 1.5954$ is parameter-free, from the embedding plus Appell–Lerch sums, with no β or tuning (l.31–33) • Unsupported in reviewed scope. No derivation, normalization, units, embedding map, or specific Appell–Lerch sum is given. It is not stated whether R_c is dimensionless or in units of GM/c^2. The only corpus hit is the compendium itself

C11 • Falsification table (l.40–42) • Absent. Only labels are present (galactic rotation curves, BH cores, strong force, 539.9 s delay). No quantitative rows or datasets. Tier E-UNTESTED; I add no data claims

C12 • "Stable core locked; parameter-free boundary verified" (l.44) • Status language only, per PROVENANCE. It is not a verification and not a lock change

Residual vs resonant: the "539 steps" in C-HQCC treats 539 as a step count. The corpus treats it as a fixed iteration budget (residual capacity), not a convergence time. These are kept separate.

5. Questions and exact proof requests (NL-numbered)

- NL-Q1 (embedding). Give the map φ from vertices of T_3 (degree 4) into $\mathbb{H}^{11}$ and prove $d_{\mathbb{H}}(\varphi(u),\varphi(v)) = c \cdot d_T(u,v)$ for all u, v. State c. If the embedding is only quasi-isometric, give the constants.

- NL-Q2 (11D / split). Write the 11D metric with signature, the (4+1)/(5+1) decomposition, and the computation showing the vacuum energies cancel. Define "negative-signature universe."

- NL-Q3 (junction). State the junction conditions across the 5D/6D interface and across the core/exterior boundary ($[K_{ij}]$, surface stress tensor). Prove them without a tuned E_{core} (inventory gap 6).

- NL-Q4 (Kummer). State precisely which ring's ideal class or factorization problem is meant ($\mathbb{Z}_3$ is a PID), define the Kronecker filter, and state what it is claimed to prove.

- NL-Q5 (curvature). Define the invariant 11D 3-adic curvature tensor: domain, value field ($\mathbb{Q}_3$ or $\mathbb{R}$), connection. Give the four projections and show they reproduce the gauge groups. Specify which framework of literature a520db70 it extends (tree curvature 1612.09580 vs formal p-adic $g_{\mu\nu}$), or that it extends neither.

- NL-Q6 (continuum). State the Tate–Gelfand limit theorem: what sequence converges to what, and in which topology, with the Archimedean metric recovered (inventory gap 3).

- NL-Q7 (singularity). Derive a bound on ρ or the Kretschmann scalar from a Written field equation. Show where the 3^{-539} floor comes from, if it does.

- NL-Q8 (HQCC). Which T_3 is meant: the supplied map (§3, divergent) or the canonical floor map? State "termination" precisely (reach 1? fixed iteration count?) and give a proof. For the supplied map, a termination proof cannot exist (§3 Theorem).

- NL-Q9 (R_c). Give the derivation of 1.5954: the exact Appell–Lerch sum (parameters, q, z), the boundary evaluation, normalization, units, and why no free parameter enters. Give enough digits to reproduce.

- NL-Q10 (falsification). Give quantitative predictions with dataset, observable, predicted value, error, and comparison baseline for each table label, and specify the 539.9 s observable.

6. Proof offers (distinct from the completed algebra)

Completed this turn: the §3 lemmas and theorem on the supplied map, and the 3-adic limit ±½.

Offers (not started; each needs an Owner-Written target statement, and invent HOLD applies):

- O1: on receipt of NL-Q1's φ, verify the isometry or distortion exactly.

- O2: on receipt of NL-Q9's sum, reproduce 1.5954 numerically and audit normalizations.

- O3: on receipt of NL-Q8's chosen map and termination notion, check it against the canonical floor map with exact arithmetic (the RPP itinerary lemma fbbac9a9 is available).

- O4: on receipt of NL-Q6's statement, compare it with Tate's thesis local factors and the Local Haar item 70.

7. Remaining blockers

- The six inventory gaps remain (b2581c1a).
- Gap 1 remains after the literature check (a520db70).
- C2–C10 need the definitions requested in NL-Q1–Q10.
- C11 has no data rows.

8. Compliance

No invented hypotheses or proofs; the §3 algebra is elementary and checked. No CERT, publishing, spending, or messages; I did not contact Gemini or any agent. Tip unchanged. Prior artifacts b2581c1a and a520db70 unchanged. Next: PP audit, then Keeper (parent handles the handoff).

3. Auditor

Substantive packet now available: 19 questions and 16 evidence asks; hash-only future verification offer, no physics evaluation. [SOURCE INTEGRITY, AUTHORITY, and claim targets explicitly labeled in Q1–Q19]. Source Q3/E2 retains an outdated PP discrepancy: PP erratum 27c8eb18 resolves it; no different-bytes copy is needed to explain that error. Source E13/E14 “open/absent” is bounded to Auditor’s consulted sources; Ops now supplies relevant specs and status records, not a security proof. Auditor header says ~14:05 CST although observed on disk at 14:03 CST; timestamp retained as source metadata, not clock verification.

Source: /workspace/quantum-comp/reports/bridges/AUDITOR/FINDINGS_AUDITOR_SOFTSF_QUESTIONS_GEMINI_COMPENDIUM_49849cae_2026-11-05.md

SHA256: e8dd86ee354cd38b28f2f08eeab1d2460f2671500d06588160a35118f9885827. Exact complete source:

FINDINGS — Auditor Soft SF questions & proof requests — Gemini/Owner compendium 49849cae… — 2026-11-05 ~14:05 CST (America/Chicago)

> Role: Auditor · REPORTS ONLY · No CERT · No RECS · invent HOLD STANDS · No publish/spend · No agent messaging.
>
> Receipt: Owner via Physicist (compendium relayed by Grok Bot 13:54 CST per PROVENANCE.md).
>
> tip 7cddb188ca340ad977516327065e8df88c3b16ed01bcd8301c5bf0f8bb5dfc91 — before MATCH / after MATCH — tip_touched NO.
>
> Authority hygiene: Source language "verified", "locked", "parameter-free", "guarantees", "Stable core locked; parameter-free boundary verified" is NOT verified authority. It is untrusted received content; it does not change any lock, lift invent HOLD, or constitute proof (PROVENANCE.md). Auditor does not evaluate any claim as true or false.

1. Source hash table

• Artifact • Expected • Found • Status

1 • references/gemini-compendium-20261105/RECEIVED_COMPENDIUM.txt • 49849cae20fe57c24533b770c13c4a53e5844f0bb75086250b9a82af0b446b73 • same • MATCH

2 • references/gemini-compendium-20261105/PROVENANCE.md • (none given) • c6fea7d234e35e96ece9114c789d32eadb41d103ddf75c487c6df56f0ad3d190 • RECORDED (matches PR2 506b2aef table)

3 • reports/OWNER_DOCUMENT_COMPREHENSIVE_COMPENDIUM_S2_11DM2ET_3ADIC_GR_EXTENSION_2026-11-05.md • b5398c04f52bad1cd1d8e5244ccce5fa2eace9d8bf2411e69851781ce2825b99 • same • MATCH

4 • reports/bridges/AUDITOR/FINDINGS_AUDITOR_SOFTSF_INVENTORY_OWNER_COMPENDIUM_S2_11DM2ET_3ADIC_2026-11-05.md • ab27db16… • ab27db163618107cfdcb31401ffb131bec7210e3ced9a4fb31afcae953009f86 • MATCH

5a • reports/FINDINGS_LOCAL_SOFTSF_LANE_REVIEW_COMPENDIUM_S2_11DM2ET_3ADIC_2026-11-05.md (Local L1–L7) • 892964f4… • 892964f496763b3073595fc13014e67892c32fa9f2dff129c3190405ad362d99 • MATCH

5b • reports/FINDINGS_PP_SOFTSF_AUDIT_LOCAL_COMPENDIUM_REVIEW_892964f4_2026-11-05.md • ddcf72ee… • ddcf72ee7b95528c28f2f849a998d89601a47ff25acf8da1806be6330fafd7ea • MATCH

6a • reports/FINDINGS_MR1_SOFTSF_GEMINI_COMPENDIUM_REVIEW_Q_PROOF_OFFERS_HOLD_INVENT_2026-11-05.md • ff177053… • ff1770535fe7aa4e157fbd205b7e1f81341f7829786e7c8dec32d6ce0c54ebeb • MATCH

6b • reports/SOFTSF_PRESS_FINDINGS_MR1_…_2026-11-05.md • 3e8007ba… • 3e8007ba6e739c5aa01f56766838b65950be80ec72e25ca327ce6911f9ac62e8 • MATCH

7a • bridges/AUDITOR/FINDINGS_PR2_SOFTSF_ACK_OWNER_MAINBOT_FALSIFICATION_LANE_S6_INVENTORY_2026-11-05.md • 506b2aef… • 506b2aef2839afeda593162b8c9192b53df2ba91286947fb4d04bb39666f7d83 • MATCH

7b • bridges/AUDITOR/SOFTSF_PRESS_FINDINGS_PR2_…_2026-11-05.md • 7c0396d9… • 7c0396d9e1bb6f39dba93a0b93b695c673c9f65d77eae15f67b402de99a470e9 • MATCH

Supporting (read, as reported by NLP):
bridges/NONLOCAL_QM_EH11D/FINDINGS_NL_COMPENDIUM_LANE_REVIEW_2026-11-05.md c9bf12b4edbd358e1d13d72cad5234df9568e01cea61862a1615f186c25613ba;
grok-files/new-HQCC-Theorem-Clarification-Topological-Derivation.txt 4d11b7a6…a91 MATCH NL cite;
reports/crypto/RPP_spec_v1/CORRECTION_3^539_3^270.md fbbac9a9…91a MATCH NL cite.

2. Auditor questions (record only; each tied to a received-text section)

Q1 (Header/PROVENANCE). Can Owner/Physicist confirm the Gemini/Google attribution is relayed only, and name any original-source artifact (file/sha) beyond RECEIVED_COMPENDIUM.txt 49849cae…?

Q2 (Closing line "Stable core locked; parameter-free boundary verified"). Confirm this line carries no lock/CERT/HOLD effect (per PROVENANCE, ab27db16, MR1 Q10).

Q3 (§3 T_3 display). PP note N1 (ddcf72ee) states the display at line 26 had numerators/exponents stripped; the on-disk 49849cae… line 26 reads "(3x+1)/3^{ν_3(3x+1)} … (3x-1)/3^{ν_3(3x-1)}". Which reading did PP audit, and is a different-bytes copy involved? Hash of the copy PP read requested.

Q4 (§3 T_3/HQCC; Local L1, NLP open check as reported). Which T_3 is meant: the map as supplied in §3, or the canonical floor map in HQCC Clarification 4d11b7a6…? (NL-Q8, recorded as reported; Auditor does not evaluate.)

Q5 (§3 HQCC theorem). Where is the HQCC statement with hypotheses and proof location (MR1 R4)?

Q6 (§1 Bruhat–Tits "isometric embedding"; Local L2, PP CONFIRM). Is the intended statement isometric or quasi-isometric? Provide map φ and constants (NL-Q1).

Q7 (§1 Dimensional splitting; Local L3). Where is the written action/junction condition behind "vacuum energy density balances perfectly"? (MR1 Q8.)

Q8 (§4 $R_c \approx 1.5954$; Local L4, PP N2). Which Appell–Lerch sum and matching function produce R_c (MR1 Q1/R1)? Is "parameter-free" recorded as author assertion only?

Q9 (§2 Scale Invariance "Tate–Gelfand pairing"; Local L5). Definition and reference for the pairing (MR1 Q3/R2)?

Q10 (§1 "9 Maths of Unification"; Local L7). Primary written source for the 9-core remainder algebra (MR1 R6)?

Q11 (§2 Purged Primes / Kronecker filter / Kummer; Local L7). Definition of the filter, list or rule for purged primes (MR1 Q4–Q5/R3)?

Q12 (§2 Unified Forces; Local L7). Which curvature-tensor components project to which forces — written source?

Q13 (§3 Singularity Prevention). Which written theorem connects the ultrametric bound to the no-singularity claim (MR1 Q9)? Local L6 records only the ultrametric inequality / Haar item 70 as Local-proved.

Q14 (§3 Gravitational Breathing Modes). Written definition location? No on-disk artifact identified by Auditor or cited in sources 4–7.

Q15 (§5 HQH-539-512). Spec and hardness-assumption artifact (Local L1 "inherits gap"; NL: Category B/open, 539 Labs lane)?

Q16 (§5 128 Logic-Qutrit Hampton Processor). Any written spec artifact? None cited in sources 4–7.

Q17 (§3 "539-step" vs §6 "539.9s"). Same object, conversion, or distinct (MR1 Q7; NL residual-capacity vs step-count separation)?

Q18 (§6 table). Table body ABSENT (PR2 506b2aef blocker 1; PROVENANCE "bracketed descriptions"). Will Owner supply the written Standard-vs-S^2 table?

Q19 (§6 Cosmic Resonance Delay 539.9s). Written prediction ± tolerance + dataset for Tier E (PR2 blocker 2)?

3. Exact proofs / evidence requested

• Artifact ask • Section • Status as found

E1 • Original-source copy of compendium (non-relayed) with sha • Header • ABSENT

E2 • Hash of the T_3 text PP read for N1 • §3 • OPEN (discrepancy vs 49849cae line 26)

E3 • Corrected / intended T_3 definition Written by Author • §3 • OPEN; canonical floor map PRESENT-on-disk 4d11b7a6… (identity to §3 map not established)

E4 • HQCC theorem statement + proof location (MR1 R4) • §3 • ABSENT

E5 • Embedding φ into $\mathbb{H}^{11}$ + constants (NL-Q1) • §1 • ABSENT

E6 • Splitting action/junction lemma (L3) • §1 • ABSENT

E7 • Appell–Lerch → R_c derivation with error bound (MR1 R1, L4) • §4 • ABSENT

E8 • Tate–Gelfand pairing definition/reference (MR1 R2, L5) • §2 • ABSENT

E9 • Kronecker filter definition vs Kummer (MR1 R3) • §2 • ABSENT

E10 • 9 Maths primary citation (MR1 R6) • §1 • ABSENT

E11 • T_3/539 → R_c / forces / V_{35} map (MR1 R5; O2 6349a521…) • §2–§4 • ABSENT/OPEN

E12 • Breathing-mode definition • §3 • ABSENT

E13 • HQH-539-512 spec + hardness assumption • §5 • OPEN (RPP correction fbbac9a9… PRESENT-on-disk as adjacent only)

E14 • Hampton Processor spec • §5 • ABSENT

E15 • §6 table body • §6 • ABSENT

E16 • 539.9s prediction ± tolerance + dataset • §6 • ABSENT (adjacent python trigger list 0ad64b5b… PRESENT, not §6 body per PR2)

PRESENT-on-disk proved piece (per Local L6): Haar item 70 (7ab0ed68…/56e59a8e…, Keeper MATCH) — ultrametric/valuation only.

4. Lane verification offer (Auditor)

Auditor offers, for any named proof artifact E1–E16 that lands: hash-only Soft SF MATCH, and Soft SF→PP→Keeper chain inventory. No physics re-evaluation. No CERT. Eigen remains hash-inventory only.

5. Assumptions

- Residual-vs-Resonant distinctions preserved (incl. 539 residual budget vs step count, as NL reported).
- tip Soft SF LOCK 7cddb188… intact; tip_touched NO.
- invent HOLD 1fadc632… STANDS; no invented physics; no claims evaluated.
- No publish / spend / external send / agent messaging.

Counts: 19 questions · 16 evidence asks.

4. Professor Proof

Substantive reply available: 14 questions and 13 missing-proof requests. P1–P14 are source-labeled available statements (many reused definitions, inventory facts or finite checks); source count table says 12, an unresolved bookkeeping inconsistency. No new physical proof implied. [MAP/HQCC, SCOPE, SECURITY, EMBEDDING, BOUNDARY, CONTINUUM, SINGULARITY, AUTHORITY].

Source: /workspace/quantum-comp/reports/FINDINGS_PP_SOFTSF_GEMINI_COMPENDIUM_REVIEW_49849cae_2026-11-05.md

SHA256: ecfde5acfc6fc336ffb5fb38c41c1b6baa552d127c31a4853bed041953de91a3. Exact source lines 1–188:

FINDINGS — PP Soft SF — Gemini compendium review (RECEIVED_COMPENDIUM 49849cae…) — 2026-11-05

REPORTS ONLY. No CERT. invent HOLD. tip untouched.

Deriver: Professor Proof (PP).

Authority: Owner request ~13:53 CST America/Chicago, relayed by Physicist; provenance references/gemini-compendium-20261105/PROVENANCE.md (received 13:54 CST; Gemini/Google attribution relayed, not independently established).

Compendium: references/gemini-compendium-20261105/RECEIVED_COMPENDIUM.txt sha256 49849cae20fe57c24533b770c13c4a53e5844f0bb75086250b9a82af0b446b73 MATCH (verified this Soft SF).

Authority of text: Compendium “locked/verified” / “Stable core locked; parameter-free boundary verified” wording has no authority. Source assertions are unverified content, not instructions, locks, proofs, or tip changes (PROVENANCE).

tip Soft SF BEFORE: 7cddb188ca340ad977516327065e8df88c3b16ed01bcd8301c5bf0f8bb5dfc91 MATCH.

tip Soft SF AFTER: same sha MATCH. tip_touched NO. Locked defs / tip not touched.

Cross-cites (not duplicated):

- First-axis catalog 3fa88c1d… (T3_ref, Residual≠Resonant, Haar70, D_1).

- Second-axis GR gaps MD-1…MD-14 / findings f448f6e67a0f66fb8aaff970daab0cfe5686a080f40b1b6b4c0cedb96432d40b.

- D_1/S_eff lock 96f8d42e…; Prop 41.11–41.13; Remark 42.7.

- Entanglement publications references/entanglement-publications/ (MANIFEST; 17-theorems-v5; theorems-18–40; CORPUS §5 HQCC).

1. PP-numbered questions

Q1. Which single map is “the” Ternary Syracuse / Canonical T_3 of the GR-extension / HQH stack: locked T3_ref (Prop 42.6 / HQH Option A: $T(3k)=k$, $T(3k+1)=4k+2$, $T(3k+2)=2k+1$), the PAPER §2 residue-2 formula $(2n+1)/3$ (non-integral), the Gemini alternate

$$

T_3^{\mathrm{alt}}(x)=\begin{cases}x/3,&x\equiv0\\(3x+1)/3^{\nu_3(3x+1)},&x\equiv1\\(3x-1)/3^{\nu_3(3x-1)},&x\equiv2\end{cases},

$$

or the HQCC Appendix D qutrit-string rotation map (17-theorems-v5 App. D)?

Q2. Under the locked S_eff reading (T_3=539 static cell/hash budget, not a stopping time; Prop 41.11; Residual≠Resonant, Prop 41.12), what Written theorem — if any — supports the compendium claim that “all interior state trajectories terminate in a bounded, 539-step convergence sequence”?

Q3. The compendium restores a “specialized Kronecker filter” and “four fundamental forces” as 3-adic tensor projections. Prop 41.13(ii) / OW lock §4 state Kronecker / four-forces / SU(3) continuum maps are PURGED from Category A. Is the compendium claiming a Cat A restoration, a Cat B narrative, or an unauthorized override?

Q4. Where is the Written definition of the “9 Maths of Unification,” the isometric Bruhat–Tits → 11D hyperbolic embedding, and the (4+1)/(5+1) signature split (compendium §1)? How do these relate to second-axis gaps MD-1 (S^2), MD-2 (11D), MD-3 (M^2ET)?

Q5. The claimed “parameter-free” boundary $R_c\approx1.5954$ numerically equals $\sqrt{2/0.7858}$ to $<4\cdot10^{-5}$. Given MD-10 ($\Lambda_0\approx0.7858$ underived), in what sense is R_c parameter-free rather than $\sqrt{2/\Lambda_0}$ for an underived Λ_0?

Q6. What is the Written statement of the “Hampton Qutrit Collatz Convergence (HQCC) theorem” that the compendium invokes for unitarity / information paradox resolution, and how does it relate to: (a) T3_ref; (b) App. D qutrit map; (c) HQCC paper 55-step / live-table 7-step variants (CORPUS §5.3)?

Q7. Compendium §5 claims HQH-539-512 security from “one-way hardness of reversing paths through the 3-adic Syracuse trajectory.” First-axis PP-C17: Π_{free} is solved by $x=3^L\cdot y$. Which Π, with which residual predicate, is alleged hard — and where is Criterion 3 (reduction) Written?

Q8. Compendium §6 lists “Cosmic Resonance Delay (539.9s)” as a falsifiable signature. Under Thm 41.10(ii) / Prop 41.11 and first-axis PP-C22, how can a temporal seconds observable be a Cat A consequence of D_1/S_eff? Is this Tier E only?

Q9. Temporal Torsion Cohomology (compendium lead): is this identified with residual form $H^2(F)\cong\mathbb{Q}\cdot[s_{\alpha\delta}]$, or a distinct undefined object? If identified, how does that square with Thm 41.10(ii) (D_1 takes no temporal argument)?

Q10. Singularity resolution (§3): against MD-4–MD-9 and MD-12–MD-14, which of {3-adic Einstein tensor, identification map, Mellin–Barnes→M_{35}, I_top, Israel junction, BH entropy/spectrum, horizon=T_3 theorem, Owner “defined BH mechanics”} does the compendium claim to supply — and where is each Written at proof level?

Q11. Tate–Gelfand pairing “continuum limit without information loss” (§2): what is the precise statement, domain, and citation? How does it close MD-5?

Q12. Entanglement Thm 29 (“Black Hole Information Resolution”) and Thms 36–40 (Einstein equations / SM / complete unification) in theorems-18-40-v1 assert Verdict: Holds. MANIFEST: documentary presence ≠ independent validation. Does the compendium treat those Verdict lines as Owner-locked proofs, or as author assertions pending audit?

Q13. Closing line “Stable core locked; parameter-free boundary verified”: which Owner Write lock sha is cited? (None appears in the received text.)

Q14. Diagram/table in the received file are bracketed descriptions only (PROVENANCE). Were authoritative figures omitted, and if so where do they live?

2. Precise missing proofs (claim → what exactly is needed)

• Claim in compendium (paraphrase) • What is needed for a proof (not offered here) • Blocks / conflicts

MP-1 • Alternate T_3^{alt} governs BH interior + HQCC 539-step termination • (i) Owner choice of one Canonical map; (ii) proof that that map's orbits terminate in ≤539 steps for the claimed domain; (iii) reconciliation with T3_ref / App. D / HQCC paper variants • Differs from T3_ref on 2/3 of ℕ; as written, $\nu_3(3x\pm1)=0$ always on residues 1,2 ⇒ map ≡ $x/3\mid 3x+1\mid 3x-1$, which grows; Python: seeds 5,7,10,539,4880 do not reach {0,1,2} in 2000 steps

MP-2 • "Universal" 539-step convergence / unitarity / information paradox resolved • Proof that residual cardinality 539 forces a dynamical stopping time — contradicts Prop 41.11 / 41.12 unless a new Owner Write overrides S_eff • Residual≠Resonant lock; 539 is static budget, not a clock

MP-3 • Kronecker filter + four forces as 3-adic projections • Cat A restoration would need Owner Write un-purging Prop 41.13(ii); else remains purged / Cat B only • OW 96f8d42e... §4 PURGED

MP-4 • Bruhat–Tits isometric embedding into 11D hyperbolic space; (4+1)/(5+1) split; vacuum-energy cancellation of Λ • Written embedding theorem; 11D metric; junction; derivation of Λ cancellation • MD-2; no such theorem in PAPER e140947e...

MP-5 • Singularity prevented by ultrametric + residual ternary • 3-adic Einstein eqs; identification map; junction; entropy/spectrum; defined BH mechanics • MD-4–MD-9, MD-12–MD-14; PAPER §4: proposal, not consequence of §§2–3

MP-6 • Gravitational breathing modes at horizon • Definition + derivation from a field equation • Undefined in PAPER / locks

MP-7 • Parameter-free $R_c\approx1.5954$ via Appell–Lerch / Bruhat–Tits • Derivation of R_c without inserting $\Lambda_0\approx0.7858$; or Owner Write deriving Λ_0 • MD-10; $R_c=\sqrt{2/\Lambda_0}$ numerically

MP-8 • Continuum limit via Tate–Gelfand pairing, no information loss • Precise pairing statement + proof of continuum match • MD-5

MP-9 • HQH-539-512 hardness from reversing Syracuse paths • Fully defined Π with residual predicate + hardness assumption not already false; Criterion 3 reduction • First-axis PP-C17–C19: Π_free easy; Criterion 3 unmet

MP-10 • Scale invariance / "9 Maths" dictating 11D splitting • Definitions of the nine maths + derivation of split • MD-1–MD-3; proprietary language ≠ theorem

MP-11 • HQCC as cited for BH unitarity • Single consistent HQCC statement + completed proof (App. D sketch ≠ proof; CORPUS: 539 vs 55 vs 70 vs 7 steps unreconciliation) • Entanglement MANIFEST: presence ≠ validation

MP-12 • Cosmic Resonance Delay 539.9 s as model signature • Cat A temporal prediction from D_1/S_eff — forbidden by Thm 41.10(ii) / Prop 41.11 unless Owner lifts lock • First-axis PP-C22; Tier E only at best

MP-13 • "Stable core locked; parameter-free boundary verified" • Cite an Owner Write sha that locks those phrases • None in received text; wording has no authority (PROVENANCE)

An offer is not a completed proof. App. D "Proof Sketch," entanglement "Verdict: Holds," and compendium "guarantees" are offers / author assertions, not PP-accepted completed proofs.

3. Proofs actually available

Status: proved / conditional / conjectural. No invented proofs.

P1. Compendium sha and provenance scope — proved

- Hypotheses: File on disk as received.

- Argument/source: sha256sum → 49849cae20fe57c24533b770c13c4a53e5844f0bb75086250b9a82af0b446b73 MATCH. PROVENANCE: no authority for “locked/verified” language; tip/locks unchanged.

- Status: proved.

P2. Locked T3_ref (not the Gemini alternate) — proved (definitional lock)

- Hypotheses: Prop 42.6 / HQH Option A / first-axis PP-C07.

- Argument/source: $T3_{\mathrm{ref}}(3k)=k$, $T3_{\mathrm{ref}}(3k+1)=4k+2$, $T3_{\mathrm{ref}}(3k+2)=2k+1$ (floor form at residue 2: $(2n-1)/3$). Integer-valued $\mathbb{N}\to\mathbb{N}$.

- Status: proved as the locked reference map. Compendium does not get to redefine it by assertion.

P3. Gemini T_3^{alt} is a different map from T3_ref — proved

- Hypotheses: Formula as printed in compendium §3.

- Argument: On residue 0 both give $x/3$. On residues 1 and 2: for all $x\equiv1\pmod3$, $3x+1\equiv1\pmod3$ so $\nu_3(3x+1)=0$; for all $x\equiv2\pmod3$, $3x-1\equiv2\pmod3$ so $\nu_3(3x-1)=0$. Hence as Written,

$$
T_3^{\mathrm{alt}}(x)=\begin{cases}x/3,&x\equiv0\\3x+1,&x\equiv1\\3x-1,&x\equiv2\end{cases}
$$

(integer-valued; no non-integer display under this reading). Agrees with T3_ref on exactly the multiples of 3 (≈1/3 of $\mathbb{N}$). Python: 3334/10000 agreement on $0\ldots9999$; differs e.g. $T3_{\mathrm{ref}}(1)=2$ vs $T_3^{\mathrm{alt}}(1)=4$.

- Status: proved.

P4. Gemini T_3^{alt} does not exhibit 539-step termination on tested seeds — proved (finite negative)

- Hypotheses: Map of P3; domain including {5,7,10,100,539,4880}.

- Argument: Iteration grows (~×3 each non-multiple-of-3 step). Python: none of those seeds reach {0,1,2} within 2000 steps (values exceed 10^{1000}). By contrast T3_ref sends 4880→1 in 14 steps (first-axis / PAPER claim under T3_ref).

- Status: proved as finite counterexamples to the compendium’s 539-step claim for this printed map. (Does not resolve Collatz-type questions for other maps.)

P5. Residual ≠ Resonant; 539 static budget — proved

- Hypotheses: Prop 41.11–41.12; OW 96f8d42e….

- Argument/source: First-axis PP-C15, PP-C20; second-axis PP-GR-C02. Compendium’s “539-step convergence” / “539.9 s delay” as dynamical/temporal universals conflict with the lock unless Owner Write overrides.

- Status: proved (lock hygiene). Compendium universal-539 dynamics conjectural / ill-typed under present locks.

P6. Ultrametric inequality — proved (standard)

- Hypotheses: $|\cdot|_3$ on $\mathbb{Q}_3$.

- Argument: Standard. Second-axis PP-GR-C05. Does not by itself yield singularity resolution (MD-4–MD-9).

- Status: proved. Physical BH use conjectural.

P7. Haar Euler factor on $\mathbb{Z}_3^4$ — proved (identity-only)

- Source: Local Soft SF 7ab0ed68…; first-axis PP-C12; PAPER §5. Compendium does not restate it; adelic/timescale narrative remains interpretive.

- Status: proved (identity); timescale reading conjectural.

P8. \(R_c\approx1.5954\) equals \(\sqrt{2/0.7858}\) — proved (numeric identity)

- Hypotheses: Λ_0=0.7858 as in PAPER §4 / prior Cosmic posts.

- Argument: \(\sqrt{2/0.7858}\approx1.595361\); abs err vs 1.5954 \(<4\cdot10^{-5}\). So “parameter-free \(R_c\)” is at best \(\sqrt{2/\Lambda_0}\) with underived Λ_0 (MD-10).

- Status: proved (equality); “parameter-free” claim conjectural / false as stated without Λ_0 derivation.

P9. Kronecker / four-forces maps are PURGED from Cat A — proved (inventory)

- Source: Prop 41.13(ii); OW lock §4; Remark 42.7; first-axis PP-C21.

- Status: proved as negative inventory. Compendium restoration conjectural / out of Cat A without new OW.

P10. Second-axis gaps MD-1…MD-14 still open; embedding/continuum/singularity claims unmet — proved (as gap status)

- Source: Findings f448f6e6…; NL inventory b2581c1a…; PAPER §4, §7. Compendium asserts resolution without supplying the missing definitions.

- Status: proved that the gaps remain; singularity-resolution package conjectural (Cat B).

P11. HQCC documentary presence ≠ completed proof — proved (process / corpus fact)

- Hypotheses: Entanglement MANIFEST + CORPUS §5 + App. D text.

- Argument: App. D states a different qutrit-rotation map and a “Proof Sketch” (Cayley graph $\mathbb{Z}_3 * \mathbb{Z}_2$, flux period 539.9 s entering as cycle length) — sketch, not a completed arithmetic proof. CORPUS §5.3/§8: HQCC step counts 539±1 vs 55 vs 70 vs live 7 steps from 4880 unreconciled; multiple maps in corpus. MANIFEST: “documentary presence is NOT independent validation of proofs.” Thms 29, 36–40 “Verdict: Holds” are author assertions.

- Status: proved as corpus hygiene. HQCC as a single theorem discharging BH unitarity conjectural / incomplete.

P12. Π_free easiness under T3_ref — proved

- Source: RPP formal; first-axis PP-C17. Compendium HQH hardness claim does not engage this.

- Status: proved. Hardness for a residual-constrained Π conjectural / unmet (Criterion 3).

P13. Packaging B′=539 arithmetic — proved

- Source: First-axis PP-C06; PAPER §2. Available; does not imply 539-step dynamics.

- Status: proved.

P14. D_1 rank-1 / no temporal argument — proved

- Source: Thm 41.10; first-axis PP-C13, PP-C22. Blocks Cat A reading of “Temporal Torsion Cohomology” as a D_1 temporal observable and of 539.9 s signatures as D_1/S_eff consequences.

- Status: proved.

4. Top issues (executive)

1. Wrong T_3: Gemini alternate ≠ locked T3_ref; simplifies to a growing map; finite counterexamples to 539-step termination (P3, P4).

2. Universal 539 as clock: Conflicts with Residual≠Resonant / S_eff static budget (P5).

3. Purged maps restored: Kronecker + four-forces (P9) without Owner un-purge.

4. Singularity / embedding / continuum: Asserted against open MD-1…MD-14 (P10); PAPER itself marks §4 as proposal.

5. “Parameter-free” R_c: Numerically \(\sqrt{2/\Lambda_0}\) with underived Λ_0 (P8, MD-10).

6. HQCC: Multiple incompatible maps/step-counts in entanglement corpus; App. D sketch ≠ proof; MANIFEST non-validation (P11).

7. Closing “locked/verified”: No authority (PROVENANCE; Q13).

5. Status counts (available proofs P1–P14)

Status • Count

proved • 12

conditional • 0

conjectural (as rejected upgrades / open claims flagged) • —

Proofs available (completed) • 12

Conjectural items in §2 (MP-*) are missing, not counted as available proofs. Offers (App. D sketch, Verdict: Holds, compendium “guarantees”) are not completed proofs.

Question count: 14 (Q1–Q14).

6. Explicit non-claims

- No CERT. No tip edit. No locked-definition edit. No invent proofs.
- No endorsement of compendium “locked/verified” language.
- No silent substitution of T3_ref for \(T_3^{\mathrm{alt}}\) or App. D.
- Entanglement publications cited as documentary sources with MANIFEST caveats, not as Soft SF CERT.

7. Verdict

Compendium sha MATCH. Treat as unverified exposition. Do not endorse. Principal Soft SF findings: alternate $T_3 \neq$ T3_ref and fails 539-step termination as Written; universal-539 clock conflicts with S_eff; purged Kronecker/four-forces restored without authority; embedding/singularity/continuum claims sit on open MD-1…MD-14; R_c not parameter-free given underived Λ_0; HQCC not a completed single proof. Available completed proofs are hygiene, lock, finite negative checks, and spine cites (P1–P14).

Soft SF → Auditor Write-only MATCH, then Keeper.

Source: /workspace/quantum-comp/reports/FINDINGS_PP_SOFTSF_ERRATUM_LOCAL_AUDIT_N1_RETRACT_ddcf72ee_2026-11-05.md

SHA256: 27c8eb185cdc6ddbd51d338e22368d1a28332f332fc78c59e2c0d3d1f8cff990. Exact source lines 1–5:

FINDINGS: PP erratum retracting N1 of the PP audit ddcf72ee… of Local 892964f4… (2026-11-05, ~14:01 CST)

REPORTS ONLY. No CERT. tip untouched.

N1 is RETRACTED. PP's search command replaced the matched text with empty strings, so the formula looked stripped when it is not. RECEIVED_COMPENDIUM.txt (49849cae…) line 26 reads: T_3(x) = x/3 if x≡0; (3x+1)/3^{ν_3(3x+1)} if x≡1; (3x−1)/3^{ν_3(3x−1)} if x≡2 (mod 3).

Local's L1 reading is therefore the source text verbatim, and L1 DEFECT is verified against the source. $\nu_3(3x\pm1) = 0$, so orbits grow and HQCC is FALSE as stated.

The same "reconstructed reading" caveat in PP's Gemini review relay is also withdrawn; the review FINDINGS ecfde5ac… stands. PP audit ddcf72ee… is otherwise unchanged: CONFIRM, with N2 only.

5. 539 Labs Ops

Substantive questions supplied by parent at 13:57 CST; completed O1 disk packet subsequently received. Six exact questions and three exact proof requests below. Source is parent steering message, not independently verified Ops transcript. [MAP/HQCC, SECURITY, PROCESSOR, BOUNDARY, OBSERVABLES].

Ops-Q1

Which T_3 is meant? The compendium’s version (L26) differs from our locked floor-$T3_{ref}$.

Ops-Q2

HQCC promises a "bounded 539-step" termination (L27). Where is the statement and proof? What does it terminate at? What is the bound relative to?

Ops-Q3

L38 says HQH-539-512 is "built entirely on 539-step convergence" and its "security relies on Syracuse path-reversal hardness". That contradicts our record: HQH is a SHA3 sandwich, Crit.3 is unmet (Close B), and the hardness is not established. Which spec is the compendium describing?

Ops-Q4

What is the basis for the "post-quantum" label (L38)?

Ops-Q5

L37 has the 128 Logic-Qutrit processor "simulating cosmological structures/brane leakage". What is the source? On our side, the LQH ALU is only a draft SV/Verilator simulation skeleton. No hardware exists.

Ops-Q6

Where do $R_c \approx 1.5954$, the Appell–Lerch matching, and 539.9 s come from?

Ops-P1

HQCC: $\forall x$ in the stated domain, the T_3 trajectory reaches a stated terminal set within $\leq B(x)$ steps, with B and the domain given explicitly.

Ops-P2

A reduction from HQH-539-512 preimage/collision resistance to a stated Syracuse-reversal problem, with advantage bounds. Note that Pre→Π is near-circular under Im(SHA3) collapse (our d14a50d7…).

Ops-P3

Any post-quantum security argument for P2.

Completed/reused evidence (parent relay, paraphrase): C1 literal map 1→4→13→40→121 diverges in ordinary absolute value because $\nu_3(3x\pm1)=0$. This targets HQCC P1, not security P2 (Ops ambiguous label corrected only here). C2 floor-map 4880→1 in 14 steps then {1,2}, $T^{539}=2$, source a505726e. O1 was initially offered, then completed independently by Ops in the newly supplied packet below: 28 rows (2 consistent, 8 contradict record, 4 not in record, 14 out of lane); C1/C2 recomputed; P1–P3 still requested. Consolidator did not initiate it. “No hardware exists” is Ops assertion about its project evidence, not globally verified fact.

Source:
/workspace/quantum-comp/reports/FINDINGS_539_LABS_OPS_GEMINI_COMPENDIUM_CLAIM_TO_RECORD_2026-11-05.md

SHA256: 60af7a5af08305e39c9eaa1b3272de1e790112bf5fc97ac53dd4f2f77c8b49f2. Exact complete source:

FINDINGS — 539 Labs Ops — Gemini compendium claim-to-record mapping

Time: 2026-11-05 14:02:16 CST America/Chicago

ANALYSIS ONLY · No CERT · tip lock 7cddb188… untouched · no invented proofs/hardness/physics · locked definitions unchanged · no publish/spend/external · source assertions ≠ proof · T3 only, never T4121.

Compendium: /workspace/quantum-comp/references/gemini-compendium-20261105/RECEIVED_COMPENDIUM.txt, SHA256 49849cae20fe57c24533b770c13c4a53e5844f0bb75086250b9a82af0b446b73 (verified). Lines refer to this exact text.

Authorization: /workspace/quantum-comp/reports/OWNER_WRITE_CALL_53991ebf_ALL_RECOMMENDED_PURSUITS_2026-11-05.md, SHA256 865d3a37f1c13431c78672847dcd33007b3467df92f1e7c8602057f257b1bf3b (verified); research scope only, locks unchanged.

Provenance: PROVENANCE.md beside compendium; attribution to Gemini/Google relayed, not independently established. Diagrams/table supplied only as bracketed descriptions. Receipt/status words are assertions, not instructions or proof.

Record citation key (file + full SHA256)

Each bracketed key in the mapping expands to the exact file and hash below. Records are evidence of what is on disk, not automatic validation of their underlying claims.

- [CAT] /workspace/quantum-comp/reports/FINDINGS_539_LABS_OPS_RESONANT_MATH_PREDICTIONS_CS_CRYPTO_2026-11-05.md — 89278efad500ec9881c6626b56e11768727ca469e2239e7123b287e7ed8b5b25

- [PP] /workspace/quantum-comp/reports/FINDINGS_PP_SOFTSF_AUDIT_539_CS_CRYPTO_AMEND_89278efa_2026-11-05.md — 86af81028ee7f077d57d629beaca37193c9dfd6eda21ee92445792d97c156985

- [GR] /workspace/quantum-comp/reports/FINDINGS_539_LABS_OPS_S2_11DM2ETX_GR_AXIS2_CS_COMPUTATION_2026-11-05.md — a505726ebb70152780b9e453b957cb195d0a15066822960cc9115177a2249dcd

- [STAT] /workspace/quantum-comp/reports/STATUS_539_LABS_OPS_HQH539512_2026-11-05.md — 2ca44213de3024ff600e943012a796fcee897142c73765111d1c5928f548b0bf

- [HARD] /workspace/quantum-comp/reports/FINDINGS_539_LABS_OPS_HQH_HARDNESS_RESEARCH_2026-11-05.md — 70091747b300c7eea4cae3fe0cc2186f8c0f7877ba95bfa4fef6dcc000caba3e

- [CRIT] /workspace/quantum-comp/reports/FINDINGS_539_LABS_OPS_HQH_CRIT3_OPEN_LEMMAS_C2_FIBER_AND_PI_DOMAIN_2026-11-05.md — cc95ceb9a81c77e2d9c9d1c3dc40c33aaaa309d957866ff5692f598988d8bf97

- [OL68] /workspace/quantum-comp/reports/FINDINGS_539_LABS_OPS_HQH_CRIT3_OL68_COMPOSITION_ADV_INSTANCE_2026-11-05.md — 72cb778f843530bf3f149791ea774da13d2cfce41f804928077cda19300f6864

- [LEDGER] /workspace/quantum-comp/reports/FINDINGS_539_LABS_OPS_HQH_CRIT3_OL12_AUTHOR_CONFIRM_AND_OL7_LEDGER_2026-11-05.md — 7835312e6926630600edaea885a9cd633077af022385e31800ad3ae0512097af

- [PINS] /workspace/quantum-comp/reports/FINDINGS_539_LABS_OPS_HQH_CRIT3_OL15_DEFINITIONAL_CHOICES_2026-11-05.md — 85ef70db16dad3b005b4a50f2ebbfe90122566c7e6f05c46e766831549ec410f

- [FRAME] /workspace/quantum-comp/reports/FINDINGS_539_LABS_OPS_HQH_CRIT3_OL8_CONCRETE_FRAMING_AND_OL11_OUTLINE_2026-11-05.md — 07521fa9a9a64be6e36ff863d2bd4f91d9f2506f489f3033449528da89f09efd

- [DRAW] /workspace/quantum-comp/reports/FINDINGS_539_LABS_OPS_HQH_CRIT3_OL3_M_DRAW_CLARIFICATION_2026-11-05.md — d14a50d797e466ec8a900be0df85913dbabf7ebe8a9d3b1f7051f216afa37433

- [ALU] /workspace/539-labs/lqh-alu/ALU_SPEC.md — f28a6b8b135d3db6e6ef228de9d408346021ff9f6d48e048e6e69aa4ace9f563

- [ARCH] /workspace/539-labs/lqh-alu/ARCHITECTURE.md — 71a0ed15ce759890514aa4336132d53c3175422c97841158b2caf3e4e383fffd

PP is the catalog amend audit, not CERT or PQ endorsement. Historical LEDGER owner-waits are superseded by PINS definitional choices and DRAW clarification only; no hardness is thereby resolved.

Claim mapping

Verdicts compare assertions to the scoped record: CONTRADICTS RECORD includes portraying explicitly unverified work as established, not merely mathematical falsity. NOT IN RECORD means unsupported in this scoped Ops mapping, not absent from every historical narrative. Physics routing is analytical assignment only, not an external transmission. Quoted fragments split compound claims.

Compendium line ref • Claim quoted • Record status with file+sha (citation key above) • Verdict • Note

L9 • A proprietary algebra based on a 9-core remainder system. • [CAT] C15; [GR] G1–G3 — No corresponding 9-core algorithm/definition in the mapped Ops record • NOT IN RECORD • 72/45 working filter is not this asserted 9-maths system; no identification inferred.

L9 • It maps discrete algebraic structures directly onto continuous macroscopic templates • [GR] G4 — No valuation-to-geometry map • OUT OF OPS LANE • Routed to GR/physics; not evaluated.

L16 • screens out incompatible prime fields using a specialized Kronecker filter routine. • [CAT] §2.2/C15 — No such routine established in this Ops catalog • NOT IN RECORD • Search hits for Kronecker products/limit formula are not an implementation or prime-purge theorem; request exact ring, inputs, predicate and preserved invariant.

L16 • This completely bypasses the Kummer ideal factorization crisis by isolating invariant ideals within Q_3. • [CAT] C15 — No supporting algorithm/theorem in mapped Ops record • NOT IN RECORD • Source assertion only; request precise ideal-theoretic statement; do not invent filter or bypass proof.

L25 • The convergence of states inside a black hole is governed by a 3-adic variant of the Collatz conjecture • [GR] F5/F7 — T3_ref is arithmetic, not spacetime dynamics • OUT OF OPS LANE • Routed to GR/physics; no physical mechanism evaluated.

L26 • x/3 if x ≡ 0 (mod 3) • [STAT] L17; [GR] F4/F5 — Same residue-zero branch as REF • CONSISTENT • Branch-only agreement, not identity of whole maps.

L26 • (3x+1)/3^{ν_3(3x+1)} if x ≡ 1 (mod 3) • [PINS] OL-1; [CAT] D1 — REF uses (4x+2)//3 • CONTRADICTS RECORD • Compendium gives 1→4; REF gives 1→2. C1 below.

L26 • (3x-1)/3^{ν_3(3x-1)} if x ≡ 2 (mod 3) • [PINS] OL-1; [GR] F4/F5 — REF uses floor((2x+1)/3) • CONTRADICTS RECORD • Compendium gives 2→5; REF gives 2→1; do not silently repair.

L27 • The Hampton Qutrit Collatz Convergence (HQCC) theorem guarantees that all interior state trajectories terminate in a bounded, 539-step step convergence sequence • [CAT] C7/C13/C16; [GR] F5; [STAT] L148 — 539 is fixed packaging; HQCC absent as crypto proof; general free termination open • CONTRADICTS RECORD • Literal-map universal boundedness disproved by C1; REF 539 schedule is not a universal stopping bound (C2). Physics interpretation excluded.

L27 • ensuring absolute unitarity and resolving the information paradox. • [GR] F5/F7 — No physical conclusion follows from arithmetic orbit • OUT OF OPS LANE • Routed to GR/physics; not evaluated.

L36 • The non-Archimedean physics of the S^2-11DM^2ET model directly translate to physical and cryptographic architectures developed by 539 Labs. • [ALU] §6; [STAT] L45–57 — Coupling is bridge-candidate only, not proven • CONTRADICTS RECORD • Asserting an established direct translation overstates a draft research bridge; physical part routed separately.

L37 • 128 Logic-Qutrit Hampton Processor • [ALU] §1/§2; [STAT] L45–57 — 128-trit engineering draft exists, not verified logical-qutrit quantum processor • CONTRADICTS RECORD • 128 classical balanced trits are not 128 demonstrated fault-tolerant logical qutrits; named code and hardware evidence missing.

L37 • Hardware designed to process ternary matrix transformations natively. • [ALU] §2; [ARCH]; [STAT] L51–55 — Listed operations are word arithmetic and tritwise logic • NOT IN RECORD • No matrix-transform native processor established by this draft; no fabricated hardware capability.

L37 • It exploits quantum information geometry to simulate deterministic cosmological structures and brane-leakage mechanics. • [STAT] L34/L54; [ALU] §6 — No continuum/brane-clock in Cat A; bridge not proved • OUT OF OPS LANE • Routed to GR/physics; no simulator/geometry mechanism evaluated.

L38 • HQH-539-512 Hash Function • [STAT] L10–28; [CAT] D2 — Specification and reference implementation exist • CONSISTENT • Existence/identity only; tests are engineering evidence, not security proof.

L38 • A post-quantum cryptographic hash • [STAT] L35/L43/L142; [CAT] mandatory non-claims — Quantum security not analyzed; proven PQ hard invent-pending • CONTRADICTS RECORD • Contradicts verified-status portrayal, not a proof that the sandwich is insecure.

L38 • built entirely on the 539-step convergence criteria. • [STAT] L16/L28; [PINS] OL-2; [CAT] C13/D2 — Public API is SHA3→fixed 539 REF steps→SHA3 • CONTRADICTS RECORD • Not entirely convergence-derived; 539 is schedule, not convergence theorem; SHA3 layers and encoding are essential.

L38 • Its security relies on the one-way hardness of reversing paths through the 3-adic Syracuse trajectory map. • [HARD]; [CRIT]; [OL68]; [LEDGER]; [PINS]; [FRAME]; [DRAW] — No completed reduction/hardness; Crit.3–4 unmet • CONTRADICTS RECORD • As established security basis this overclaims. Π_free easy, bounded variant heuristically easy; SHA3-image/fiber issues open. Not a sandwich insecurity proof.

L3 • resolves long-standing cosmological singularities • [GR] §0/F7/G1–G7; [STAT] process locks — No Ops adjudication of physical assertion • OUT OF OPS LANE • Singularity resolution: routed to GR/physics lanes for review only; not evaluated or sent externally.

L3/L17 • unifies the four fundamental forces as 3-adic tensor projections • [GR] §0/F7/G1–G7; [STAT] process locks — No Ops adjudication of physical assertion • OUT OF OPS LANE • Force unification: routed to GR/physics lanes for review only; not evaluated or sent externally.

L3/L18 • guarantees scale invariance without manual parameter tuning • [GR] §0/F7/G1–G7; [STAT] process locks — No Ops adjudication of physical assertion • OUT OF OPS LANE • Scale invariance/Tate–Gelfand continuum assertion: routed to GR/physics lanes for review only; not evaluated or sent externally.

L7 • isometric embedding of the Bruhat-Tits tree into an 11-dimensional hyperbolic space. • [GR] §0/F7/G1–G7; [STAT] process locks — No Ops adjudication of physical assertion • OUT OF OPS LANE • Embedding: routed to GR/physics lanes for review only; not evaluated or sent externally.

L8 • Spacetime intervals are evaluated using the 3-adic valuation • [GR] §0/F7/G1–G7; [STAT] process locks — No Ops adjudication of physical assertion • OUT OF OPS LANE • Spacetime metric interpretation (not the standalone ultrametric inequality): routed to GR/physics lanes for review only; not evaluated or sent externally.

L10 • Vacuum energy density balances perfectly across this junction to eliminate the cosmological constant problem. • [GR] §0/F7/G1–G7; [STAT] process locks — No Ops adjudication of physical assertion • OUT OF OPS LANE • Dimensional split/vacuum cancellation: routed to GR/physics lanes for review only; not evaluated or sent externally.

L22–24 • preventing a true mathematical singularity (R=0) from forming. • [GR] §0/F7/G1–G7; [STAT] process locks — No Ops adjudication of physical assertion • OUT OF OPS LANE • Singularities and gravitational breathing modes: routed to GR/physics lanes for review only; not evaluated or sent externally.

L31–32 • $R_c \approx 1.5954$ • [GR] §0/F7/G1–G7; [STAT] process locks — No Ops adjudication of physical assertion • OUT OF OPS LANE • Appell–Lerch/parameter-free matching and exact GR transition: routed to GR/physics lanes for review only; not evaluated or sent externally.

L42 • Cosmic Resonance Delay (539.9s) • [GR] §0/F7/G1–G7; [STAT] process locks — No Ops adjudication of physical assertion • OUT OF OPS LANE • Cosmic observables, rotation curves, black-hole cores, strong force; only bracketed table supplied: routed to GR/physics lanes for review only; not evaluated or sent externally.

L44 • Stable core locked; parameter-free boundary verified. Holding next-step options for Owner. • [GR] §0/F7/G1–G7; [STAT] process locks — No Ops adjudication of physical assertion • OUT OF OPS LANE • Final status assertion is not authority or a lock change: routed to GR/physics lanes for review only; not evaluated or sent externally.

COMPLETED — independently executed Python checks

C1 — Literal L26 map does not divide on nonzero branches

[COMPLETED] Python 3 executed in this review, using integer arithmetic and a valuation loop. For $x \equiv 1$, $3x+1 \equiv 1 \bmod 3$; for $x \equiv 2$, $3x-1 \equiv 2 \bmod 3$. Thus $\nu_3 = 0$ on both nonzero branches. Starting at 1, every iterate remains $\equiv 1$ and follows $a_{k+1} = 3a_k + 1$. Executed list (k=0…10):

[1, 4, 13, 40, 121, 364, 1093, 3280, 9841, 29524, 88573].

Closed form $a_k=(3^{(k+1)}-1)/2$ was asserted against all eleven computed values. The recurrence proves unbounded divergence in the usual real/integer magnitude, not a statement about 3-adic convergence. This refutes the literal universal bounded-integer/539-step claim; it does not repair L26 or adjudicate black-hole physics.

C2 — Floor-T3_ref fixed-round record value

[COMPLETED] [GR] F4/F5 explicitly states the floor rule, so recomputation is permitted: r0 n//3; r1 (4n+2)//3; r2 (2n+1)//3. Independently executed 539 iterations from 4880:

4880→3253→4338→1446→482→321→107→71→47→31→42→14→9→3→1 (14 steps).

Then 1↔2; 539−14=525 is odd. Python assertions passed: step 14 = 1 and T3_ref^539(4880)=2. This reproduces [GR] SHA256 a505726ebb70152780b9e453b957cb195d0a15066822960cc9115177a2249dcd F5, including its cited PP N1 confirmation. Finite computation is not a universal convergence or security theorem. C1 and C2 concern different maps.

Reproduction code actually executed (equivalent compact form):

```
python

def v3(n):

k = 0

while n % 3 == 0:

n //= 3

k += 1

return k

def literal(x):

if x % 3 == 0: return x // 3

y = 3*x + (1 if x % 3 == 1 else -1)

return y // 3v3(y)

a = [1]

for _ in range(10): a.append(literal(a[-1]))

assert a == [(3(k+1)-1)//2 for k in range(11)]

def ref(n):

return n//3 if n%3 == 0 else (4*n+2)//3 if n%3 == 1 else (2*n+1)//3

b = [4880]

for _ in range(539): b.append(ref(b[-1]))

assert b[14] == 1 and b[539] == 2
```

OFFERED / REQUESTED — NOT COMPLETED proofs

- P1 HQCC bound statement + proof — REQUESTED, not done. Supply exact domain, map, quantifiers, meaning of convergence/termination and a bound. A universal bounded integer theorem for literal L26 cannot survive C1; any different statement must explicitly distinguish itself without changing locked REF definitions. Local Ops offer: audit the supplied statement against the fixed-round/free-orbit distinction; no HQCC proof offered as accomplished.

- P2 HQH-to-Syracuse-reversal reduction with advantage bounds — REQUESTED, not done. Need explicit R for the exact H_ref Pre/SPR/Col game, target domain/distribution, composition model and proven advantage relation. OL-6/7/8 remain invent-blocked; ε_C2 blank, OL-9 hardness not minted; Crit.3 unmet and Crit.4 review external. DRAW separates m drawn from an unspecified D_M, SHA3-derived seed support, and solution domain D=[0,2^512); neither D_M nor its entropy is silently supplied here. FRAME fixes concrete (539,512,512), not an asymptotic security family. Offer: review a supplied reduction and its losses, not manufacture one. Definitions and structural collision partitions are not hardness.

- P3 post-quantum argument — REQUESTED, not done. Need a specified quantum adversary/oracle model, security target, explicit assumptions and concrete query/advantage accounting for the full sandwich. No Shor/Grover immunity, complexity-slice membership or proven PQ follows from ternary arithmetic, KATs, 539 rounds, C1 or C2. Offer: audit supplied assumptions/argument only.

Physics routing / boundaries

R_c≈1.5954 and Appell–Lerch matching, 539.9 s observables, singularity/information-paradox claims, forces, spacetime metric, embedding, dimensional split, brane leakage and scale invariance are routed analytically to GR/physics lanes without evaluation. No physical proof, experiment, field equation or valuation-to-geometry bridge invented. The 3-adic norm inequality alone is not proof of the claimed spacetime construction. No external message sent.

Tier E remains separate: recorded ALU simulation/KAT/RTL/avalanche results are engineering evidence only, not re-run here and not promoted to security or physics proof. Only C1/C2 were independently computed for this deliverable.

File discipline

Only the requested new FINDINGS, its byte-identical reports copy and their new SHA256 sidecars are deliverables. Exclusive-create mode prevents overwrite. Existing-file content hashes are compared to a pre-write baseline; no existing record is edited, renamed or overwritten. Tip content remains untouched. No CERT, publishing, spending or external action.

Mapping row counts: CONSISTENT=2, CONTRADICTS RECORD=8, NOT IN RECORD=4, OUT OF OPS LANE=14; total=28.

6. Agent

Substantive reply available: 15 questions, 12 requests, 16 source-labeled offers (3 documentary, 10 already-proved-from-source, 3 conditional). Reused proofs are not new proofs completed in this consolidation. Full question/request/offer text and assumptions retained. [MAP/HQCC, EMBEDDING, SPLIT/VACUUM, FORCES, FILTER/KUMMER, CONTINUUM, BOUNDARY, SECURITY, SCOPE]. Stale /workspace/reports paper copies are outside this compendium scope; no edits made.

Source: /workspace/quantum-comp/reports/FINDINGS_AGENT_GEMINI_COMPENDIUM_RUNDOWN_QUESTIONS_PROOFS_OFFERS_2026-11-05.md

SHA256: b83f95f865266d9e6e59970debe30032555844b3531d6a42456e9ccc99674d73. Exact source lines 68–202:

Critical T3 map CONFLICT (must-call-out)

Axis • Claim / definition • Soft SF status

Gemini §3 • $T_3(x)=x/3$ if $x\equiv0$; $((3x\pm1)/3^{\nu_3(3x\pm1)})$ if $x\equiv1,2$; HQCC “guarantees” all interior trajectories terminate in a bounded 539-step convergence; absolute unitarity / information paradox “resolved” • CONFLICT / OVERCLAIM vs corpus

Option A / REF / STAT / post-floor [GR] • $T_3(3k)=k$, $T_3(3k+1)=4k+2$, $T_3(3k+2)=\lfloor(2n+1)/3\rfloor$ (≡ $2k+1$)); 539 = frozen schedule/budget, not free-orbit guarantee • MATCH anchors [A][REF][STAT][GR][V]

Free orbit (floor) • $4880\to1$ in 14 steps; cycle $\{1,2\}$; frozen 539 iterations from 4880 terminal 2 (not halt-at-1) • proved [V] / [AX2] G8

Entanglement PDFs • Yet another HQCC Map Definition: $3n$ / $((3n+1)/2$ rotation rules (SUMMARY reverify; v5 p.117). Different from both Gemini §3 and Option A floor map • documentary caveat — do not invent reconciliation

Agent does not invent a bridge among these three T3 presentations.

Comparison table — Gemini claim → corpus status

• Gemini claim (paraphrase) • Corpus status • Notes

C1 • 11D isometric Bruhat–Tits embedding into hyperbolic space • UNDEFINED / OVERCLAIM • No Soft SF-written isometric chart in [GR]/[AX2] U2

C2 • Dimensional split (4+1)/(5+1); vacuum-energy balance eliminates Λ • UNDEFINED / OVERCLAIM • No signature/partition lemma Soft SF-admissible; [AX2] U2

C3 • Four forces = projections of 11D 3-adic curvature • UNDEFINED / OVERCLAIM • Unestablished physical bridge; refuse as theorem

C4 • Kronecker filter completely bypasses Kummer crisis • UNDEFINED / OVERCLAIM • No Soft SF definition of that filter in Agent corpus

C5 • Tate–Gelfand continuum limit without weight-level collapse • UNDEFINED / OVERCLAIM • Classical Tate/Gelfand neighbors ≠ model continuum theorem

C6 • Ultrametric prevents true singularity $R=0$ • OVERCLAIM vs [AX2] G10–G11,G17 • Ultrametricity alone ≠ positive floor; positivity ≠ no outer pole; full GR extension conjectural

C7 • Gravitational breathing modes at horizon • UNDEFINED / conjectural • Not a Soft SF-proved object in Agent extracts

C8 • T3 map as Gemini §3 $((3x\pm1)/3^\nu)$ + HQCC 539-step guarantee • CONFLICT • See Critical T3 map above

C9 • Parameter-free $R_c\approx1.5954$ via Appell–Lerch • UNDEFINED / OVERCLAIM • No Soft SF-checked evaluation in Agent anchors

C10 • Cosmic Resonance Delay 539.9 s • OVERCLAIM / type error • Conflates schedule integer with seconds; residual≠resonant; no Soft SF step→seconds map

C11 • HQH-539-512 security from one-way hardness of reversing Syracuse paths • OVERCLAIM • Option A freezes verification semantics; [RES] A3/A4/GAP7 — no hardness reduction Soft SF-proved

C12 • 128 Logic-Qutrit Hampton Processor / brane-leakage • UNDEFINED (hardware/marketing) • Outside Soft SF-admissible math spine

C13 • Closing: “Stable core locked; parameter-free boundary verified” • OVERCLAIM (authority hygiene) • Not Soft SF lock/CERT; invent HOLD STANDS; tip untouched

C14 • Residual Ternary Algebra + Resonant Algebra as unified physical continuum replacement • CONFLICT on identification • [AX2] G5 / [RES] A1: residual539≠resonant539 proved separation

C15 • Entanglement pubs / HQCC as Soft SF proof of unitarity • OVERCLAIM • Documentary only [EM]; third T3 map; no invented reconciliation

MATCH rows (limited, non-physical): ultrametric inequality statement on $\mathbb{Q}_3$ is standard; integer 539 appears as residual capacity and resonant schedule in [GR]/[RTA]; Haar Euler factor exists as local identity under stated hypotheses — none of these upgrade Gemini physical/unification claims.

A. QUESTIONS (Agent) — numbered

1. T3 map identity. Soft SF asks: which of (i) Gemini §3 $((3x\pm1)/3^{\nu}$ branches, (ii) Option A / REF floor map $T_3(3k)=k$, $T_3(3k+1)=4k+2$, $T_3(3k+2)=\lfloor(2n+1)/3\rfloor$, (iii) entanglement-PDF HQCC Map $3n$ / $(3n+1)/2$, is claimed as the Soft SF-admissible T_3 for S²-11DM²ET? They are not Soft SF-identical; which written source reconciles them without inventing a bridge?

2. HQCC as theorem. Where is Hampton Qutrit Collatz Convergence stated with hypotheses and proof Soft SF can audit? Soft SF currently has no Agent Soft SF MATCH packet validating HQCC as proved. How does it relate to RTA Prop.42.6 Category A facts without inventing a map?

3. 539-step “guarantee.” Soft SF asks: is “539-step interior convergence” meant as (a) free-orbit termination bound, (b) frozen Option A schedule of exactly 539 iterations, or (c) something else? Under (b), Soft SF [V] proves terminal value 2 from seed 4880, not halt-at-1. Which reading does Gemini §3 assert?

4. 539 vs 539.9 s. Soft SF asks for the Soft SF-written conversion (if any) from resonant schedule count 539 to “Cosmic Resonance Delay 539.9 s.” Absent that map, Soft SF treats the seconds claim as type-error / OVERCLAIM.

5. Bruhat–Tits isometric embedding. Into which explicit 11D hyperbolic model, with which isometric chart, is the tree embedded when boundary matching is performed? Soft SF finds no Agent-checked derivation.

6. Signature split (4+1)/(5+1). Where is the Soft SF-written partition/signature lemma forcing this split and “perfect” vacuum-energy balance eliminating Λ?

7. Forces-as-projections. Which curvature-tensor components, on which domain, project to which four forces? Soft SF asks for a written intertwining map, not marketing prose.

8. Kronecker filter / Kummer bypass. Exact definition of the filter; which ideals of which ring; how does it “completely bypass” Kummer factorization rather than rename an unrelated 3-adic screen?

9. Tate–Gelfand continuum limit. Precise bilinear pairing, spaces, and theorem preventing “weight-level collapse” for this model (not classical neighbor names alone)?

10. $R_c\approx1.5954$. Which Appell–Lerch sum (indices, characteristics, nome/modulus, regularization) evaluates to this number without free parameters, and where is the Soft SF-readable evaluation?

11. Singularity prevention as physics theorem. Which Soft SF theorem, under which stress-energy ansatz, states ultrametric bounds prevent $R=0$ for collapsing cores? Soft SF [AX2] G10 shows $3^k\to0$ in $\mathbb{Q}_3$; G11 shows positivity of inner radius does not rule out outer g_{rr} pole.

12. Gravitational breathing modes. Soft SF-admissible definition and derivation location? Presently UNDEFINED in Agent extracts.

13. HQH security. Soft SF asks: which Soft SF-proved reduction equates “reversing 3-adic Syracuse trajectories” with HQH-539-512 security? Option A [A] freezes reference semantics and explicitly does not complete a reduction ([RES] A3/A4).

14. Authority hygiene. Soft SF asks Owner/Physicist to confirm: Gemini closing “Stable core locked; parameter-free boundary verified” does not lift invent HOLD, tip LOCK, or authorize CERT / publish / spend / Gemini send — MATCH [PRV].

15. Entanglement PDF T3 vs paper T3. Soft SF asks whether deposited pubs’ HQCC Map is intended as (a) alternate definition, (b) distinct object, or (c) erratum candidate — Soft SF will not invent reconciliation ([SUM] reverify).

B. EXACT PROOFS REQUESTED (Agent) — what would be needed (not invented)

• Requested Soft SF-admissible object • Why needed for Gemini claim

P1 • Written statement+proof (or Soft SF-auditable packet) of HQCC with hypotheses; explicit relation/non-relation to RTA Prop.42.6 • Gemini §3 unitarity / information-paradox claim

P2 • Proof that free $\mathbb{N}$-orbits under the canonical floor T_3 all enter $\{1,2\}$ (or another stated attractor) and separately, if claimed, that a 539-step bound is a theorem rather than Option A freeze • “539-step guarantee” vs schedule

P3 • Explicit Soft SF definition equating or mapping Gemini §3 branches ↔ Option A floor ↔ entanglement HQCC Map — or a Soft SF declaration they are distinct • T3 CONFLICT

P4 • Derivation: named Appell–Lerch evaluation → $R_c\approx1.5954$ with error bound / parameter inventory • §4 boundary claim

P5 • Definition+proof of “Tate–Gelfand” continuum limit for S^2-11DM^2ET • §2 scale-invariance

P6 • Definition of Kronecker filter + precise theorem vs Kummer factorization • §2 purged primes

P7 • Signature/partition lemma for (4+1)/(5+1) and vacuum-energy cancellation of Λ • §1 dimensional splitting

P8 • Intertwining map: 11D 3-adic curvature → four forces • §2 unified forces

P9 • Isometric Bruhat–Tits embedding chart into stated 11D hyperbolic model used at the boundary • §§1,4

P10 • Soft SF step-count → seconds map (if any) for 539.9 s, with units and observational protocol • §6 Cosmic Resonance Delay

P11 • Cryptographic reduction for HQH-539-512 (or Soft SF statement that none is claimed) • §5 cryptography

P12 • Valuation→geometry + junction + conservation constructions discharging [AX2] U8–U12 / G17 before singularity-resolution is Soft SF-admissible as theorem • §§3–4 singularity / BH mechanics

Until P1–P12 exist as Soft SF-readable sources, corresponding Gemini assertions remain ABSENT / OVERCLAIM / CONFLICT relative to Agent corpus — not Soft SF CERT (Agent issues No CERT).

C. AVAILABLE PROOF / OFFERS (Agent) — only landed or source-grounded

Status key: proved | conditional | conjectural | documentary. Assumptions / source path+sha / status required.

C.1 Documentary offers

1. OFFER / documentary — T3 residue-2 floor display now corrected on 20 GR Markdown copies (Owner-authorized). Source: [LED] 84eeb544…, [SUM] 5b395a16…; successor [GR] bd39eb4c…, [GRpin] 0009e735…. Tip untouched. Agent VERIFY [V] 08f22ba2… previously recommended; Agent did not apply.

2. OFFER / documentary — Entanglement pubs present with MANIFEST caveats; PDFs unchanged by floor fix; third HQCC Map noted. Sources: [EM] 6480105e…, [E1]/[E2] hashes above, [SUM] reverify. Status: documentary (presence ≠ proof).

3. OFFER / documentary — Stale /workspace/reports/ paper+pins still at predecessor hashes; live /workspace/quantum-comp/ copies successor. No silent rewrite this Soft SF.

C.2 Proved-from-source offers (Agent Soft SF already landed / consequence rows)

4. OFFER / proved — residual539 ≠ resonant539 (typed separation; equality of integers supplies no chain/schedule map). Sources: [AX2] G5; [RES] A1; [RTA] Props.41.11–41.12 1fe7b8a2…; [L] 96f8d42e….

5. OFFER / proved — D1 is rank-one thin-F classifier; no temporal/metric argument in the locked domain. Sources: [AX2] G4; [RTA] Thm.41.10; [L].

6. OFFER / proved — S_{eff} capacity ≠ clock; no forced $\Omega\equiv0\pmod{539}$. Sources: [AX2] G5; [RES] A2; [L] decision 2.

7. OFFER / proved — Canonical $\mathbb{Z}_3$ T_3 (branch form) is Haar-preserving and conjugate to the one-sided ternary shift under RTA Prop.42.6(iii) hypotheses. Source: [AX2] G9; [RTA] 1fe7b8a2…. Not a BH dissipation model.

8. OFFER / proved — Haar Euler factor identity on $\mathbb{Z}_3^4$ under stated max-norm / Re$((\nu+3)>0$ hypotheses. Sources: [AX2] G14; [RES] A11; paper §5. Not a physical timescale.

9. OFFER / proved — Option A reference tuple freeze (canonical branches; exactly 539 rounds; REF finalize). Source: [A] 93cc1f32…; [RES] A3/A4. Verification semantics only — not hardness.

10. OFFER / proved — Floor-map free orbit $4880\to1$ in 14 steps; cycle $\{1,2\}$; 539 frozen iterations from 4880 terminal 2. Sources: [V] 08f22ba2…; [AX2] G8; [REF] fef5fce5…; [STAT] 2ca44213….

11. OFFER / proved — Prior exact-division residue-2 display was nonintegral (hygiene defect); now display-corrected on ledger paths. Sources: [V] G7 table; [LED].

12. OFFER / proved — Ultrametricity alone does not impose positive floor 3^{-539}; $3^k\to0$ in $\mathbb{Q}_3$. Source: [AX2] G10.

13. OFFER / proved — Name-level underdetermination of S² / 11D / M²ET as coupled GR model from examined sources. Source: [AX2] G1.

C.3 Conditional offers (explicit assumptions)

14. OFFER / conditional — If 539 is independently identified with ternary resolution depth, then proposed floor $\varepsilon=3^{-539}$. Assumption: extra depth reading in [GR] §3–4; not forced by S_{eff}. Source: [AX2] G6.

15. OFFER / conditional — Free-depth vs forced-budget distinction under ATTACH attractor-depth premise (~14). Assumption: ATTACH §2 premise; not a universal orbit theorem. Source: [RES] A5.

16. OFFER / conditional — Layer2 classifier packaging unique up to $\mathbb{Q}^\times$ under PP-CHOSEN L2-D1 and $H^2(F)=\mathbb{Q}\cdot[s_{\alpha\delta}]$. Assumption: PP definition. Source: [RES] A9. Not physical transfer.

C.4 Explicit REFUSALS (would require unestablished physical bridges / invent)

Agent Soft SF REFUSES to offer as Soft SF-admissible proofs or lane completions:

- Forces unification / four-force projections as theorems.

- Singularity resolution as a physics theorem from ultrametricity alone.

- Black-hole information paradox resolution via HQCC marketing prose.

- Cosmic observables (539.9 s, galactic rotation, falsification table) as Soft SF-proved.

- Hardware / crypto security proofs for Hampton Processor or HQH-539-512.

- Invented Bruhat–Tits embedding, $(4+1)/(5+1)$ split, vacuum-Λ cancellation, Kronecker–Kummer bypass, Tate–Gelfand continuum theorem, Appell–Lerch→R_c, or reconciliation of the three T3 maps.

- Treating Gemini “verified/locked” as Owner Soft SF / tip LOCK / CERT.

Gaps Gemini text pretends closed (Agent U1–U14 / A-axis)

From [AX2] U1–U14 (still open; floor-display U6 hygiene item display-amended on ledger paths, free-vs-frozen distinction still required):

- U1–U3: S² / 11D / M²ET definitions absent as coupled model.

- U4: coupling maps residual/T3 → GR / modular law vectors absent (RTA Remark 42.7).

- U5: capacity→depth / units; residual≠resonant; no D1 clock.

- U6: residue-2 display now floor-corrected on 20 copies; free orbit vs 539 schedule distinction remains.

- U7: averaging measure / global convergence proof absent.

- U8–U12: valuation→geometry, M35/l_top, core tensors/conservation, junction/mass/entropy, basins/horizon/time law — open.

- U13: 11D↔ternary modular merger incomplete (Bridge PARTIAL / READY NO).

- U14: entanglement pubs do not discharge bridges.

From [RES] GAP list: no residual↔resonant identification; no universal free-orbit termination theorem Soft SF-extracted; filter 45/72 unresolved as proofs; Bridge C open; no Bio/Neuro/Cosmo grounding; Option A ≠ hardness.

7. Eigenvalues and matrix

Substantive reply available: 14 questions, 8 requests, 19 mixed-status rows, not 19 completed proofs. Toy identities, numerical checks, false/conjectural/conditional entries retain their source labels. [FILTER/KUMMER, FORCES, MAP/HQCC, SECURITY, PROCESSOR, EMBEDDING, CONTINUUM, BOUNDARY, SCOPE]. Parent warns chat typo x≢1 mod3; use source's nonzero residue branches instead. Hankel assertions derive from adjacent corpus, not an explicit Hankel matrix in received L38.

Source: /workspace/quantum-comp/reports/FINDINGS_EIGEN_GEMINI_COMPENDIUM_REVIEW_2026-11-05.md

SHA256: 1ae82aee3b55db1386428969cb9fc0033c866b49aa229fec09ebf79aa56355a2. Exact source lines 1–134:

FINDINGS — Eigenvalues and matrix lane: Gemini/Owner S²-11DM²ET compendium review — 2026-11-05 ~13:57 CST America/Chicago

REPORTS ONLY. No CERT. Nothing sent anywhere (incl. Gemini). No invented geometry or proofs.

Lane: linear algebra / spectral theory / matrices / ζ_{49} CRT / N36 / Q89 lifts / circulant·Hankel·det.

Residual ≠ Resonant (kept separate throughout).

Compendium status assertions ("Stable core locked; parameter-free boundary verified") recorded as author assertions, not authority.

Integrity

object • sha256 • note

RECEIVED_COMPENDIUM.txt • 49849cae20fe57c24533b770c13c4a53e5844f0bb75086250b9a82af0b446b73 • MATCH expected

PROVENANCE.md • c6fea7d234e35e96ece9114c789d32eadb41d103ddf75c487c6df56f0ad3d190 • Owner live call 13:53 CST; Gemini attribution relayed, unverified

tip BEFORE • 7cddb188ca340ad977516327065e8df88c3b16ed01bcd8301c5bf0f8bb5dfc91 • /workspace/residual-c-field-WRITTEN_PREMISES_DRAFT.md

tip AFTER • (footer) • tip_touched NO

Provenance (verbatim gist): received 2026-11-05 13:54 CST from Grok Bot as full text; source assertions unverified; review requested by Owner via live call; distribute questions / proof requests / lane proof offers; no publishing/spending or invented proofs.

Sources (re-hashed this turn)

- S_comp RECEIVED_COMPENDIUM.txt — 49849cae…

- S_prov PROVENANCE.md — c6fea7d2…

- S1 FINDINGS_EIGEN_CAT_A_FOOLPROOF_539_CORPUS_S21_2026-11-03.md — 755ec712…

- S2 FINDINGS_EIGEN_CAT_A_FOLD_LABS539_CORE_REPORTS_ONLY_2026-11-03.md — 29b6f6e2…

- S3 FINDINGS_EIGEN_SOFTSF_LOOSEN_CAT_B_11D_TERNARY_2026-11-05.md — deebdc5e…

- S4 FINDINGS_EIGEN_SOFTSF_O5_N36_O6_LANE_RESOLUTION_2026-11-05.md — 3af326ba…

- S5 EIGEN-TE-RHOBAR-N36-CRT36Z-NAMED-GAP-DIAG_2026-10-27.md — eca39084…

- S6 EIGEN-TE-RHOBAR-N36-E1-P7952701-MATCH_2026-10-27.md — 9cac9a39…

- S7 EIGEN-TE-RHOBAR-Q89-ZETA49-SOLVE42G-COMPLETE_42of42_2026-10-27.md — d4aa7ea1…

- S8 539_Labs_Technical_Corpus.md — 5a8f8630… (§2.1 lines 19–28; Hankel/HQCC line 17; Resonant §3 line 33+)

- S9 PROOF_HANKEL_POSITIVITY.md — 663f6c4f…

- Compute: grok-batches/eigen-gemini-compendium-review-2026-11-05/compute/ (checks.py, compute.log, SHA256SUMS)

Separation rule (Eigen)

- Residual Ternary Algebra = circulant group-ring $\mathbb{Z}[\mathbb{Z}/3]$ / §2.1 path-sum (W,N), det ρ(h), spr — in-lane, disk-backed (S1,S8).

- Resonant Algebra (compendium §2; corpus §3 “Gluon Resonant Algebra”) = undefined proprietary object in Eigen sources; not identified with residual circulant. No merge.

(A) Numbered questions — attributed to Eigenvalues and matrix

1. [Eigenvalues and matrix] What is the precise matrix (or endomorphism of a named module) realizing the “specialized Kronecker filter routine” that “screens out incompatible prime fields” and “isolates invariant ideals within $\mathbb{Q}_3$”? What are its eigenvalues, and over which ring is it defined? (Compendium §2 “Purged Primes”.)

2. [Eigenvalues and matrix] For the claimed “distinct projections of an invariant 11D 3-adic curvature tensor” that become the four forces: write each projection operator P_i. Do they satisfy $P_i^2 = P_i$, $P_i P_j = 0$ ($i \neq j$), and $\Sigma P_i = I$ on a named space? (Compendium §2 “Unified Forces”.)

3. [Eigenvalues and matrix] Is “Resonant Algebra” the same as, or an extension of, the residual circulant algebra $\mathbb{Z}[\mathbb{Z}/3]$ of corpus §2.1 ($h_1=[0]+[1]$, $h_2=-[0]+[1]+3[2]$)? If different, give the multiplication table / generators. Do not identify residual with resonant without a Written isomorphism. (Compendium §2 vs §3; S8 lines 25–27 vs 33–35.)

4. [Eigenvalues and matrix] The Ternary Syracuse map is written $T_3(x)=(3x\pm1)/3^{\nu_3(3x\pm1)}$ for $x\equiv\pm1 \pmod 3$. For such x, $3x\pm1 \not\equiv 0 \pmod 3$, so $\nu_3=0$ and $T_3(x)=3x\pm1$. Was this the intended formula? If not, supply the corrected map whose orbits are claimed to be 539-step bounded. (Compendium §3 HQCC; compute C2.)

5. [Eigenvalues and matrix] HQCC asserts every interior trajectory terminates in a bounded 539-step sequence. Under the map as written, seeds 1..100 yield zero cycles in 539 steps and monotone growth (compute C2). Which hypotheses (domain, map correction, 3-adic topology) restore the claim?

6. [Eigenvalues and matrix] HQH-539-512 is said to use “multi-layered Hankel matrices.” Give at least one explicit Hankel matrix H (entries) used by the hash. Is security claimed from $\det H \neq 0$, from PSD (all eigenvalues ≥ 0), or from something else? (Compendium §5; S8 line 17; S9.)

7. [Eigenvalues and matrix] The “128 Logic-Qutrit Hampton Processor” processes “ternary matrix transformations.” Is Π_fast the averaging projector of corpus/Codex, or the engineering stub eye(128) (S2 line 23)? If the former, write Π_fast and confirm $\Pi_{fast}^2 = \Pi_{fast}$ and its spectrum {0,1}.

8. [Eigenvalues and matrix] Does the compendium claim to produce, or to depend on, a characteristic-0 Kr matrix over $\mathbb{Q}$ (or $\mathbb{Q}(\zeta_{49})$) for N36? Eigen’s 12-/13-prime CRT reconstruction remains a named gap (ok 0). Clarify: no silent closure of the N36 Kr-QQ gap. (S4,S5,S6.)

9. [Eigenvalues and matrix] Q89 ζ_{49}: Eigen has mod-p landing dim/rank 89 at p=1027606441 for all 42 roots (S7). Does any compendium object require a char-0 lift of that basis? If yes, state the embedding $\mathbb{Q}(\zeta_{49})\hookrightarrow\mathbb{C}$ (or $\mathbb{Q}_p$) used.

10. [Eigenvalues and matrix] “9 Maths of Unification / 9-core remainder system”: is this circulant algebra of length M=9 (eigenvalues $h(\omega^j)$, $\omega=e^{2\pi i/9}$), a remainder system mod 9, or something else? Give the 9×9 (or other) matrix if spectral.

11. [Eigenvalues and matrix] “Isometric embedding of the Bruhat–Tits tree into 11D hyperbolic space”: if this is meant as a Hilbert-/ℓ^2-isometry ($U^*U=I$), supply the operator. If it is metric-geometry only, Eigen has no matrix to check; confirm the claim is out of Eigen scope. (Compendium §1, §4.)

12. [Eigenvalues and matrix] “Tate–Gelfand pairing” continuum limit: name the pairing as a bilinear form / matrix of pairings on named spaces, and state the spectral condition that prevents “weight-level collapse.” (Compendium §2 “Scale Invariance”.)

13. [Eigenvalues and matrix] R_c ≈ 1.5954 from “specialized Appell–Lerch sums”: is R_c an eigenvalue, a spectral radius, a determinant ratio, or a non-spectral matching constant? Give the exact expression before decimal truncation.

14. [Eigenvalues and matrix] Closing assertion “Stable core locked; parameter-free boundary verified”: which algebraic object (matrix, ideal, RREF, kernel dimension) is “locked,” and what Soft SF hash / computation verifies it? Eigen currently holds N36 Kr QQ / zero-polar / G1-mem as REMAINING (S4).

Count (A): 14

(B) Exact proofs requested (statement + why)

1. Corrected T_3 + HQCC (or withdraw).

Statement needed: Either (i) a Written map T_3': $\mathbb{Z}\to\mathbb{Z}$ (or $\mathbb{Z}_3\to\mathbb{Z}_3$) such that every orbit enters a finite set of size ≤539 within ≤539 steps, with proof; or (ii) withdrawal of HQCC as stated for the printed T_3.

Why: For printed T_3, if $x\equiv 1 \pmod 3$ then $3x+1\equiv 1 \pmod 3 \Rightarrow \nu_3=0 \Rightarrow T_3(x)=3x+1$ (likewise $x\equiv 2$). Eigen compute C2: seeds 1..100 have 0 cycles in 539 steps and grow. HQH-539-512 one-wayness inherits the gap. Compendium asserts HQCC as theorem — that is an author assertion until repaired.

2. Force-projection calculus.

Statement needed: On a named free module M (e.g. $(\mathbb{Z}/3)^n$ or $\mathbb{Q}_3^n$), operators $P_1..P_4$ with $P_i^2=P_i$, orthogonal, complete, obtained from a Written curvature tensor R by a Written formula $P_i = f_i(R)$.

Why: “Forces = projections of 11D 3-adic curvature” is undefined as linear algebra until the P_i exist; Eigen can then check $P^2=P$ and spectra $\{0,1\}$.

3. Kronecker filter as an endomorphism.

Statement needed: An explicit matrix/Kronecker construction K (or family K_p) and a theorem: “p is purged ⇔ $\lambda=0$ is an eigenvalue of K_p” (or equivalent ideal-theoretic statement with a spectral translation).

Why: “Bypasses Kummer ideal factorization crisis” is not a matrix claim Eigen can verify; the only related PROVED fact is the standard Kronecker-product eigenvalue identity (compute C5), which does not purge primes.

4. Hankel data for HQH-539-512.

Statement needed: Explicit Hankel H and a theorem that the security reduction uses PSD / nonnegative spectrum (not mere $\det\neq 0$).

Why: S9 PROOF_HANKEL_POSITIVITY: H=[[1,2],[2,1]] has det=−3≠0 but eigenvalue −1 (ghost). Compendium’s Hankel narrative supplies no matrix (S1,S3#5).

5. Residual↔Resonant bridge (or permanent separation).

Statement needed: Either a Written isomorphism/functor relating Resonant Algebra to residual $\mathbb{Z}[\mathbb{Z}/3]$ circulants preserving det/spr/W/N, or an explicit Owner note that Resonant claims are out of Eigen residual scope.

Why: Compendium §2 Resonant vs §3 Residual Ternary are juxtaposed; Eigen forbids silent identification (lane rule; S3).

6. N36 Kr over $\mathbb{Q}(\zeta_{49})$ (or confirm ABSENT).

Statement needed: Existence of a coherent Φ_{49} embedding and char-0 Kr with named height/field, or confirmation that the compendium does not claim this.

Why: S4/S5: 12-prime and Q(√−7) holdout ok 0/3000; named gap STANDS. Compendium must not be read as closing it.

7. Π_fast spectrum.

Statement needed: Formula for Π_fast on $(\mathbb{C}^3)^{\otimes k}$ or $\mathbb{C}^{128}$ with proof $\Pi^2=\Pi$ and eigenvalues in $\{0,1\}$, rank stated.

Why: S2 coded stub is Id (trivially idempotent) and does not prove the corpus averaging operator.

8. R_c exact spectral (or non-spectral) expression.

Statement needed: Exact closed form for R_c (before ≈1.5954) and whether it is an eigenvalue/spr/det ratio of a Written matching matrix.

Why: Decimal alone is not an Eigen object; “parameter-free verified” is an author assertion (compendium close).

Count (B): 8

(C) Lane-supported proofs / offers

• Offer / claim • Hypotheses • Source (path + line) • Status

C1 • Ultrametric inequality \ • x+y\ • _3 ≤ max(\ • x\ • _3,\ • y\ • _3) on ℚ • x,y∈ℚ (samples; standard on $\mathbb{Q}_3$) • S_comp L8; compute C1 • PROVED (standard; verified on samples)

C2 • Printed T_3: for x≡1 (mod 3), $T_3(x)=3x+1$; for x≡2, $T_3(x)=3x-1$ • T_3 as written in S_comp L26 • S_comp L26; compute C2 • PROVED (algebraic identity)

C3 • HQCC “all trajectories terminate in bounded 539-step sequence” for printed T_3 • Printed T_3; domain ⊇{1..100} • S_comp L27; compute C2 • FALSE (0/100 cycles in 539; monotone growth)

C4 • Residual §2.1: det ρ(h_1)=2, det ρ(h_2)=36; target [1] ⇒ W=0, N=2 • Group ring $\mathbb{Z}[\mathbb{Z}/3]$; target [1] inferred reading • S8 L25–27; S1 L16–23; compute C3 • PROVED (with stated reading; residual only)

C5 • \ • W\ • ≤ N ≤ ∏\ • h_i\ • _1 on §2.1 example; ∏\ • h_i\ • _1 = spr(ρ(\ • T\ •)) for nonnegative circulants • Residual circulants; Perron • S8 L21; S1 L25–26 • PROVED on example; general spr identity standard for nonnegative circulants

C6 • \ • W\ • ≤ ∏ spr(ρ(h_i)) in general • Residual • S8 L22; S1 L27; S3#1 • CONJECTURAL (MATCH-on-example only; invent-pending)

C7 • det≠0 does not prevent path cancellation (residual) • §2.1 over ℤ • S8 L27; S1 L24 • PROVED (det 2·36=72≠0, W=0)

C8 • Corpus “odd M ⇒ det ρ≠0” over ℤ/3 for h_2 • M=3, h_2 • S1 L28; compute C3 • FALSE over ℤ/3 (det 36≡0 mod 3); PROVED over ℤ/ℂ for h=1+x type

C9 • Avg projector $P=11^T/3$ on $\mathbb{C}^3$: $P^2=P$, eigs {1,0,0} • Standard LA • compute C4 • PROVED (toy; not claimed as Π_fast)

C10 • Π_fast = eye(128) stub is idempotent • labs539_core stub • S2 L23, L30 • PROVED as stub only; corpus averaging formula CONJECTURAL/ABSENT

C11 • DFT_3 is unitary ($F^HF=I$) numerically • Standard • compute C4 • PROVED numerically (err~1e−51); exact sympy simplify inconclusive

C12 • No Bruhat–Tits / 11D isometry matrix in compendium ⇒ U*U=I unchecked • — • S_comp L7,L31; compute C4 • CONJECTURAL/ABSENT as Eigen matrix claim (geometry → Local; Eigen does not invent)

C13 • Kronecker product: eig(A⊗B)={λμ} • Standard LA • compute C5 • PROVED; does not prove “Kronecker filter purges primes” (ABSENT/NEEDS-DEFINITION)

C14 • Hankel det≠0 ⇏ no negative eigenvalue / no ghost • H=[[1,2],[2,1]] • S9; S1 L35–36; compute C6 • PROVED category fact; compendium Hankel security claim CONJECTURAL/ABSENT data

C15 • (4+1)+(5+1)=11 arithmetic • — • S_comp L10; compute C7 • PROVED (trivial); vacuum-balance claim CONJECTURAL/ABSENT derivation

C16 • N36 Kr over ℚ / ℚ(√−7) via CRT • 12–13 e1 primes • S4; S5; S6 • CONJECTURAL named gap REMAINING (holdout ok 0); mod-p rank 503 PROVED at listed primes

C17 • Q89 ζ_{49} char-0 basis lift • coherent embedding • S7; S3#8; S4 • CONDITIONAL: mod-p rank 89 at p=1027606441 for 42/42 roots PROVED; char-0 lift CONJECTURAL

C18 • Resonant Algebra / Tate–Gelfand / 9 Maths / Temporal Torsion as Eigen matrices • — • S_comp §§1–2; S8 §3 • CONJECTURAL/NEEDS-DEFINITION in Eigen lane (not residual)

C19 • Closing “Stable core locked; parameter-free boundary verified” • — • S_comp L44 • Author assertion only — not an Eigen Soft SF status; N36/zero-polar/G1-mem still REMAINING (S4)

Count (C): 19 (status mix as tagged)

Compute summary (light sympy only; Sage N36 left running)

See eigen-gemini-compendium-review-2026-11-05/compute/compute.log. Highlights: ultrametric OK; T_3 defect PROVED; HQCC FALSE on 1..100; residual W=0,N=2 MATCH; $P^2=P$ OK for avg projector; Kronecker eig identity OK; Hankel ghost counterexample OK.

Not done / out of scope

- No tip edit; no CERT; nothing sent to Gemini or elsewhere.
- No invented Bruhat–Tits / hyperbolic / Appell–Lerch geometry proofs (Local lane owns those blockers).
- Residual path-sum ≠ Resonant Algebra: no merge.
- Long Sage N36 e1 sequence left untouched.

Tip AFTER: 7cddb188ca340ad977516327065e8df88c3b16ed01bcd8301c5bf0f8bb5dfc91 — tip_touched NO. No CERT.

8. Math Research 1

Substantive reply available: 10 questions, 6 exact requests, 5 already-supplied citation packets and offers 6–10. Status update: 6+9+10 COMPLETED as citation/numerical forensic (literature neighbors 4, project evaluation 0), not a new R_c proof. Offers 7+8 STARTED per parent; no completion supplied. [BOUNDARY, EMBEDDING, CONTINUUM, FILTER/KUMMER, MAP/HQCC, SCOPE].

Source: /workspace/quantum-comp/reports/FINDINGS_MR1_SOFTSF_GEMINI_COMPENDIUM_REVIEW_Q_PROOF_OFFERS_HOLD_INVENT_2026-11-05.md

SHA256: ff1770535fe7aa4e157fbd205b7e1f81341f7829786e7c8dec32d6ce0c54ebeb. Exact source lines 1–87:

Soft SF FINDINGS — MR1 Gemini-compendium review: questions / proof requests / citation-only offers — 2026-11-05 ~13:56 CST

Soft SF Write-only → Physicist (Owner-requested) + Auditor. REPORTS ONLY. invent HOLD STANDS.

tip Soft-SF LOCK tip_pre 7cddb188ca340ad977516327065e8df88c3b16ed01bcd8301c5bf0f8bb5dfc91 tip_touched NO.

No invent/prove BEGIN; No CERT; No publish/spend; No external sends; No RECS.

Content language “verified / locked / parameter-free” in the received text is not Soft SF authority and does not change locks.

Soft SF source integrity

Object • sha256 / note

references/gemini-compendium-20261105/RECEIVED_COMPENDIUM.txt • 49849cae20fe57c24533b770c13c4a53e5844f0bb75086250b9a82af0b446b73 MATCH Owner-relayed

PROVENANCE.md • Received 2026-11-05 ~13:54 CST via Grok Bot; Gemini attribution relayed, not independently established

invent HOLD reclaim Soft SF • 1fadc632… STANDS

PRIMARY RULE • bcc3ced0… LIVE

Resonant catalog Soft SF • 7a0be1d5… (O1–O4 all ABSENT/OPEN after cite deepen; ROLLUP d29a5ed6…)

tip • 7cddb188… tip_touched NO

MR1 lane = citation / literature Soft SF under invent HOLD. Publication assertions ≠ Soft SF proof validation (same hygiene as entanglement pubs Soft SF 32b454cc…).

A. MR1-attributed numbered questions (for Owner / Physicist)

Focus claims: Appell-Lerch · Tate–Gelfand · Kummer · Kronecker · boundary $R_c \approx 1.5954$.

1. Appell–Lerch → $R_c \approx 1.5954$. Which exact Appell–Lerch sum (indices, characteristics, elliptic modulus / nome, regularization) is claimed to evaluate to $R_c\approx 1.5954$? Where is the written evaluation (page/lemma) that produces this number without free parameters?

2. Isometric Bruhat–Tits embedding. Into which model of 11D hyperbolic space, with which explicit isometric chart, is the Bruhat–Tits tree embedded when the boundary matching is performed? Soft SF finds no MR1-checked derivation of this embedding in the received text.

3. Tate–Gelfand pairing as continuum limit. What is the precise bilinear pairing written as “Tate–Gelfand,” on which spaces (test functions / distributions on $\mathbb{Q}_3$ or adelic), and which theorem states it prevents “weight-level collapse”? Classical Tate local zeta / Gelfand–Graev / Bruhat–Schwartz references do not automatically supply the claimed continuum limit for S²-11DM²ET.

4. Kronecker filter vs Kummer crisis. What is the exact definition of the “specialized Kronecker filter routine,” and which ideals of which ring are “invariant in $\mathbb{Q}_3$”? How does this “completely bypass” the Kummer ideal-factorization problem (classically about cyclotomic integers / unique factorization failure), rather than renaming an unrelated 3-adic screen?

5. Purged primes. Which primes are purged, by which decision procedure, and what Soft SF-checkable invariant is preserved? Is there a written list or generating rule?

6. HQCC theorem. Where is the Hampton Qutrit Collatz Convergence theorem stated with hypotheses and proof (not restated in marketing prose)? Soft SF currently has no MR1 Soft SF MATCH packet validating HQCC as proved. Relation to v16 Prop.42.6 (Category A facts on T_3 / 539 only) must be spelled out without inventing a map to g_s, C, V_{35} (O2 Soft SF 6349a521… ABSENT/OPEN).

7. 539-step vs 539.9 s. The text conflates a “539-step convergence sequence” with cosmic “539.9 s” resonance delay in §6. Are these the same Soft SF object, a derived conversion, or distinct claims? What Soft SF citation binds step-count to seconds?

8. Dimensional splitting (4+1)+(5+1). Where is the Soft SF-written signature/partition lemma that forces this split and vacuum-energy balance “perfectly,” eliminating Λ? Which curvature tensor components are projected to the four forces?

9. Ultra-metric singularity prevention. Which theorem states that the 3-adic ultra-metric bound prevents $R=0$ for collapsing cores under the model’s stress-energy ansatz (not merely the ultra-metric inequality on $\mathbb{Q}_3$)?

10. Authority hygiene. Soft SF asks Owner/Physicist to confirm: Gemini/rundown “Stable core locked; parameter-free boundary verified” does not lift invent HOLD, tip LOCK, or authorize CERT / publish / spend — MATCH PROVENANCE.md.

B. Exact proofs / citations Soft SF requests (inputs needed)

• Requested object • Why Soft SF needs it

R1 • Written derivation: named Appell–Lerch sum(s) → numerical $R_c\approx 1.5954$ with error bound • Boundary claim is the sharp falsifiable number in §4; absent formula ⇒ claim ABSENT for Soft SF

R2 • Definition + proof or Soft SF-auditable reference for “Tate–Gelfand pairing” continuum limit in this model • §2 scale-invariance claim

R3 • Definition of Kronecker filter + precise statement vs Kummer factorization • §2 purged-primes / Kummer bypass

R4 • HQCC statement, hypotheses, proof location (Zenodo DOI / tip lemma / Soft SF packet) • §3 unitarity / information-paradox claim

R5 • Explicit map or theorem relating T_3 / 539 packaging to R_c, forces-as-projections, or Hilbert V_{35} objects • Align with Soft SF O2 ABSENT (no written $T_3\to g_s,C,V_{35}$ map)

R6 • Primary citation for “9 Maths of Unification” remainder algebra (not proprietary paraphrase) • §1 backbone claim

Until R1–R6 exist as Soft SF-readable sources, Soft SF status for those Gemini assertions = ABSENT / unverified content (not NO-MATCH against a written project theorem Soft SF has already certified — Soft SF has No CERT).

C. What MR1 citation-only lane can actually supply or offer now

Status key: OFFER = Soft SF Write-only invent-free work MR1 can start under HOLD; SUPPLY = already Soft SF-shipped; CANNOT = would require invent/prove BEGIN or tip edit / CERT.

C.1 Already SUPPLY (cite Soft SF packets; do not relabel as Gemini-verified)

1. SUPPLY — T_3 / 539 Category A packaging facts as written in RTA v16 Prop.42.6 (partition $3\cdot 68+5\cdot 67$; equidistribution; Haar-preserving + conjugate to shift on $\mathbb{Z}_3$): catalog + O2 Soft SF 6349a521…. Does not supply HQCC, nor a map to g_s,C,V_{35} (O2 remains ABSENT/OPEN).

2. SUPPLY — Residual ≠ Resonant packaging constraints Prop.41.11–41.12 (O2 packet).

3. SUPPLY — Hilbert $\mathbb{Q}(\sqrt{35})$ disc-140 / LMFDB dim-14 MATCH Soft SF 06c0fca1… (Thm 41.7 neighbor). Does not identify Gemini force-projections or R_c.

4. SUPPLY — O1–O4 Soft SF cite deepen ROLLUP d29a5ed6…: Ξ/Petersson, $T_3\to$ modular targets, boundary of C vs V_{35}^{Hilb}, QMF of scalar C/g_s all ABSENT/OPEN. Useful negative inventory if Gemini implies those gaps are closed.

5. SUPPLY — Entanglement pubs ACCESS Soft SF 32b454cc… with hygiene: publication assertions ≠ Soft SF proof validation.

C.2 Soft SF OFFER (invent-free; citation / livecheck / extract only)

6. OFFER — Soft SF Write-only Appell–Lerch literature matrix vs claimed $R_c\approx 1.5954$: open classical Appell–Lerch / mock-Jacobi / Zwegers completion sources; report whether any evaluated constant matches 1.5954 under stated hypotheses; no invented identification of the project sum. Status after offer: PRESENT neighbors / ABSENT project evaluation unless R1 appears.

7. OFFER — Soft SF Write-only Tate local-zeta / Gelfand–Graev / Bruhat–Schwartz citation pack clarifying classical “Tate” / “Gelfand” pairings; mark Gemini continuum-limit claim ABSENT until R2.

8. OFFER — Soft SF Write-only Kummer / Kronecker classical citation pack (Kummer on cyclotomic factorization; Kronecker limit formula / Kronecker symbol — distinguish which “Kronecker” is meant); mark “filter bypasses Kummer crisis” ABSENT until R3.

9. OFFER — Soft SF numerical forensic only (no proof): recompute $2\ln(6+\sqrt{35})/\sqrt{35}$ and other v16 Prop.41.5 constants; check whether 1.5954 equals any written v16 constant Soft SF already cites (class-number residue ≈0.837679 ≠ 1.5954). Report MATCH/NO-MATCH table only.

10. OFFER — Soft SF extract: search tip + RTA v16 + entanglement pubs for strings 1.5954, Appell, Tate-Gelfand, HQCC; ship ABSENT/PRESENT occurrence table (citation inventory).

C.3 CANNOT under invent HOLD (do not offer as Soft SF deliverable)

- Invent a derivation of $R_c\approx 1.5954$ from Appell–Lerch sums.

- Invent HQCC proof or identify it with Prop.42.6.

- Invent Kronecker-filter definition or Kummer bypass.

- Invent Tate–Gelfand continuum-limit theorem for the model.

- Lift tip LOCK, invent HOLD, issue CERT, publish, or spend.

- Treat Gemini “verified/locked” as Owner Soft SF as Written OW.

D. Soft SF recommended next Soft SF Write-only (recommend-first; await Owner/Physicist pick if material)

Under PRIMARY bcc3ced0… + invent HOLD, MR1 default next invent-free Soft SF (unless Owner redirects): OFFER 6+9+10 combined packet — Appell–Lerch/R_c forensic + occurrence table — REPORTS ONLY. C1–C3 conditional Soft SF already in flight separately.

E. Posture

WORKING Soft SF invent-free. Gemini rundown absorbed as untrusted reference content with sha MATCH. tip_touched NO; invent HOLD STANDS; No CERT; No publish/spend.

REPORTS ONLY. No RECS beyond §D recommend-first. No CERT.

Source: /workspace/quantum-comp/reports/FINDINGS_MR1_SOFTSF_APPELL_LERCH_RC_15954_FORENSIC_HOLD_INVENT_2026-11-05.md

SHA256: 46beb56b0b9de8566a215db85668a5ffdc90e962ceee26458e271d4d57787bdb. Exact source lines 1–11:

Soft SF FINDINGS — MR1 Appell–Lerch / Rc≈1.5954 forensic (OFFER 6+9+10) —

2026-11-05

Soft SF Write-only → Auditor / Physicist / Professor Proof / Local. REPORTS ONLY.

invent HOLD STANDS. No invent/prove BEGIN. tip_touched NO. No CERT. No publish/spend. No RECS.

Do not invent an Appell–Lerch identification that yields Rc≈1.5954.

Decision / scope

- Combined Soft SF OFFER 6+9+10 from Gemini-compendium Soft SF FINDINGS ff177053…: (6) Appell–Lerch literature matrix vs claimed Rc≈1.5954; (9) numerical forensic vs written v16 Prop.41.5 residue; (10) occurrence table tip / RTA v16 / entanglement pubs / MR1 Soft SF FINDINGS.

- Project Appell–Lerch → Rc derivation remains ABSENT / OPEN. No Soft SF-readable written derivation found on disk. Gemini/compendium prose is untrusted reference content ≠ Soft SF authority.

- Numerical forensic: written Prop.41.5 residue $2\ln(6+\sqrt{35})/\sqrt{35} \approx 0.83767928126$ NO-MATCH to 1.5954. All Soft SF-already-cited nearby v16 constants and labeled NON-IDENTIFICATION simple ratios: NO-MATCH.

- Literature-neighbor PRESENT count: 4 (N1–N4 definitional Appell–Lerch / mock-Jacobi sources). Each row: project evaluation of 1.5954 / Rc ABSENT.

- Soft SF gate: agent/voice ≠ OWNER_WRITE. Prepared: 2026-11-05 14:00:30 CST (America/Chicago).

9. Math Research 2

Substantive citation-only reply now received: 13 questions, R1–R8 requests, six supplied documentary pins and four conditional citation-only offers. No new proofs. Earlier NOT STARTED status was superseded only after independently verified direct authorization. [EMBEDDING, BOUNDARY, SPLIT/VACUUM, CONTINUUM, SINGULARITY, MAP/HQCC, FORCES, AUTHORITY]. Not-found statements concern two deposited texts, not global nonexistence.

Source: /workspace/quantum-comp/reports/MR2_GEMINI_COMPENDIUM_CITATION_REVIEW_2026-11-05.md

SHA256: 6f773883ab6deefb387cfcab88dea4bf42bded30dcacf58cf323310adc28aa98. Exact complete source:

MR2 — Gemini compendium citation-only review — 2026-11-05 ~14:01 CST (America/Chicago)

REPORTS ONLY. No CERT. invent HOLD / all locks unchanged. tip untouched. No invented proofs. No publish / spend / external sends / git mutations. Contribution file (not a Page).

Lane: MR2 Soft SF = geometric / non-Archimedean / 11D-compactification citation inventory under invent HOLD (prior Cat A Soft SF posture).

Authority: Owner live-call resume of MR2 citation-only Gemini compendium review (Physicist transcript ~pos 16394, Nov 5 ~13:58 CST America/Chicago); independently verified in that original user message as relayed to this executor. Source assertions in the received text are not Soft SF authority.

Focus (this pass only): embedding · 11D splitting · junction · continuum · singularity. Other literature / prediction tasks remain queued.

0. Integrity

Object • sha256 • Note

references/gemini-compendium-20261105/RECEIVED_COMPENDIUM.txt • 49849cae20fe57c24533b770c13c4a53e5844f0bb75086250b9a82af0b446b73 • MATCH expected

references/gemini-compendium-20261105/PROVENANCE.md • c6fea7d234e35e96ece9114c789d32eadb41d103ddf75c487c6df56f0ad3d190 • Received 13:54 CST; Gemini attribution relayed, not independently established; review requested Owner live call ~13:53 CST

references/entanglement-publications/MANIFEST.md • 6480105e6efa4b08b04817aabf03b57188397f55bef552c94d7c4190644b0786 • Coverage ≠ proof validation; heading/body mismatches flagged

references/entanglement-publications/17-theorems-v5.txt • b203c940b0ef81a369dc0b985400302028404070f85355f82d8eb3403761a98c • DOI 10.5281/zenodo.18442276 (v5)

references/entanglement-publications/theorems-18-40-v1.txt • e3615b32ab04b1958d99fb0d7266ec82e58dc06aca80ac19be46fc7363a44ba3 • DOI 10.5281/zenodo.18509384 (v1)

references/entanglement-publications/coverage.csv • b6c7e76559b152d4ca561ee29ce62bcd29d43b584bf255303a83af9070d7ce7f • Headings 1–40 present

tip /workspace/residual-c-field-WRITTEN_PREMISES_DRAFT.md BEFORE • 7cddb188ca340ad977516327065e8df88c3b16ed01bcd8301c5bf0f8bb5dfc91 • LOCK

tip AFTER • 7cddb188ca340ad977516327065e8df88c3b16ed01bcd8301c5bf0f8bb5dfc91 • tip_touched NO

Hygiene (MANIFEST + prior Soft SF): documentary presence / “Verdict: Holds” / “parameter-free” / “guarantees” = author assertions, not Soft SF proof validation. Heading/body mismatches in 17-theorems-v5 (T5–T11, T13–T17; T17 heading mass-range vs body β_PBH fraction) matter for citation pins. Do not fill missing definitions.

1. Focus-claim inventory (compendium → sources)

Status key: existing theorem = named classical result that exists independently of the model (does not verify model hypotheses); conditional = model claim that would follow *if* stated hypotheses were written and held; unproved = asserted as theorem/proof in project pubs but only sketch / unreconciliation / author verdict; missing definition = object used without Soft SF-checkable definition in consulted sources.

ID • Target claim • Compendium location • Hypotheses implied • Source in entanglement pubs / elsewhere consulted • Applicable fields • Status

C1 • Isometric embedding of Bruhat–Tits tree into 11D hyperbolic space • §1 L7; §4 L31 • Named BT tree over $\mathbb{Q}_3$ (or which local field?); named model of $\mathbb{H}^{11}$ (or Lorentzian?); explicit isometry • ABSENT as string/claim in 17-theorems-v5.txt, theorems-18-40-v1.txt, MANIFEST. Zenodo refs list “Hyperbolic Measure Theory” (record 17877353) by title only — PDF not on disk this pass • p-adic geometry; hyperbolic geometry; AdS/tree lit (generic) • missing definition of the embedding as model object; generic BT-tree / p-adic AdS theorems are existing theorem neighbors only — do not verify the model chart

C2 • 11D partitioned into (4+1)D negative-signature + (5+1)D positive-signature; vacuum energy balances “perfectly” across junction; eliminates cosmological-constant problem • §1 L10 • Signature split; junction hypersurface; vacuum-energy matching condition; projection of Λ • Entanglement intro 17-theorems-v5.txt L148–150, glossary L5834–5844: (4+1)D −U and (3+1)D +U (not 5+1). Dimensional arithmetic in pubs leaves remaining dims as compactification language, not a written (5+1) macroscopic sector. Vacuum-energy independence of τ_{ent} appears as T8–T10 (coverage.csv rows 9–11; text L1660+, L1739+) — independence of delay ≠ vacuum-balance / Λ-cancellation proof • GR / extra-dim / signature geometry • missing definition for (5+1) positive sector as written in Gemini text; conflict with deposited (3+1)+U; junction “perfect balance” unproved / missing definition; T8–T10 are author-asserted delay-independence statements (and sit in heading/body offset region per MANIFEST) — not Soft SF validation of Λ-elimination

C3 • Continuum limit via “Tate–Gelfand pairing”; no weight-level collapse / information loss • §2 L18 • Named pairing spaces; continuum topology; weight filtration • ABSENT “Tate-Gelfand” / “Tate–Gelfand” in entanglement txts. Nearest continuum language: T35 proof step 4 “Continuum Bit-lattice $N\to\infty$)” (theorems-18-40-v1.txt L227–228) — different object • local Tate zeta / Bruhat–Schwartz / Gelfand–Graev (generic) • missing definition of model “Tate–Gelfand”; classical Tate / Gelfand pairings = existing theorem neighbors only

C4 • Ultra-metric 3-adic bound prevents true singularity $R=0$ for collapsing stellar cores • §3 L23 • Stress-energy ansatz; meaning of R (curvature scalar? areal radius?); Planck-scale restriction map • Entanglement singularity language differs: intro L374 “11D terms smooth singularities”; T21 L78–79 “Singularity resolution” via $\int E_{\mathrm{bind}}\,dt<\infty$ (Gabriel’s Horn analogy); T37 L246 “Discrete $N_{\mathrm{simplex}}=1001$ prevents singularities”. No ultra-metric $R=0$ theorem matching Gemini wording • GR singularity theorems (Penrose–Hawking) are existing classical results about continuum Lorentzian manifolds — they do not certify the 3-adic claim • Gemini ultra-metric singularity claim = missing definition + unproved under model hypotheses; deposited pubs offer alternate singularity narratives (unproved / proof-sketch level)

C5 • Gravitational breathing modes at horizon; discrete oscillating boundary phases • §3 L24 • Horizon coordinate chart; mode spectrum • Intro L171: $G_4=539.90\pm 0.05\,\mathrm{s}$ “gravitational breathing mode”; also L233. Horizon-as-breathing-mode mechanics not derived in consulted focus passages • — • Object G_4 present as named period (author assertion); horizon breathing-mode derivation missing definition / unproved

C6 • HQCC guarantees all interior BH trajectories terminate in bounded 539-step sequence → unitarity / information paradox resolved • §3 L25–27 (printed T_3) • Domain of T_3; relation of printed arithmetic T_3 to App. D qutrit map; BH state → integer/qutrit encoding • App. D 17-theorems-v5.txt L6192–6225: different map (qutrit rotation rules); labelled “Proof Sketch”; classical Collatz remains open (L6229–6230). T29 BH information (theorems-18-40-v1.txt L154–163): leakage to −U / Page curve — not the printed Gemini T_3. CORPUS §5 / §8: 539±1 vs 55 vs 70 step unreconciliation • dynamical systems / 3-adic Collatz • HQCC as Soft SF-proved theorem = unproved (sketch + unreconciliation). Printed Gemini T_3 vs App. D map = missing definition bridge. T29 unitarity = author assertion (“Verdict: Holds”), not Soft SF CERT. Generic Collatz = existing open problem — not a model verification

C7 • Parameter-free transition boundary $R_c\approx 1.5954$ from isometric BT embedding + “specialized Appell–Lerch sums” • §4 L31–32 • Named Appell–Lerch indices; regularization; link to BT embedding • ABSENT 1.5954, Appell, Appell-Lerch in entanglement txts consulted • mock modular / Appell–Lerch (generic) • missing definition of the project sum; numerical claim unproved here; classical Appell–Lerch existence = existing theorem neighbor only

C8 • Four forces = projections of invariant 11D 3-adic curvature tensor • §2 L17 • Named curvature tensor; projection operators • T40 embedding language (theorems-18-40-v1.txt L283–284): Strong D2 3-cycles; EW 11D gauge reduction; EM

U(1) projection; Gravity leakage curvature — not “3-adic curvature tensor projections” • — • missing definition for Gemini’s 3-adic curvature-projection calculus; T40 force-embedding = separate author narrative (unproved as Soft SF)

C9 • “9 Maths of Unification” / 9-core remainder system dictates 11D geometric splitting • §1 L9 • Algebra generators; remainder map; splitting functor • ABSENT as defined object in entanglement txts; Zenodo title list includes “Resonant Number Theory” etc. — PDFs not opened this pass • — • missing definition

Cross-cut: Compendium closing L44 “Stable core locked; parameter-free boundary verified” has no Soft SF authority (PROVENANCE). Does not lift invent HOLD / tip LOCK / authorize CERT.

2. MR2-attributed numbered questions (for consolidated PDF)

1. [MR2 · Embedding] Which Bruhat–Tits tree (over which local field, which apartment system) is embedded, and into which precise model of 11-dimensional hyperbolic (or Lorentzian) space? Cite the written isometric chart (lemma/page). Soft SF finds ABSENT in RECEIVED_COMPENDIUM and in deposited entanglement txts (17-theorems-v5.txt, theorems-18-40-v1.txt).

2. [MR2 · Embedding / R_c] Which Appell–Lerch sum (indices, characteristics, nome) evaluates to $R_c\approx 1.5954$ under the claimed isometric boundary matching? Where is the evaluation written? Soft SF: strings 1.5954 / Appell ABSENT in entanglement references consulted.

3. [MR2 · 11D splitting] Compendium §1 asserts (4+1) negative-signature + (5+1) positive-signature. Deposited entanglement intro/glossary assert (4+1) −U + (3+1) +U (17-theorems-v5.txt L150, L5836–5840). Which Written split is operative, and where is the signature/partition lemma?

4. [MR2 · Junction] What is the junction hypersurface (equation, induced metric, Israel-type or p-adic matching conditions), and which identity forces vacuum-energy densities to "balance perfectly" so that Λ is eliminated? Distinguish this from T8–T10 delay-independence of global vacuum energy (coverage.csv Thms 8–10; MANIFEST: those body texts sit in heading-offset region).

5. [MR2 · Continuum] Define the "Tate–Gelfand pairing" as used in §2: domain, codomain, bilinearity, and the precise theorem that continuum limit avoids "weight-level collapse." Soft SF: name ABSENT in entanglement txts; do not silently substitute classical Tate local zeta or Gelfand–Graev.

6. [MR2 · Continuum vs T35] Is T35's "Continuum Bit-lattice $N\to\infty$" (theorems-18-40-v1.txt L227) intended as the same continuum limit as Gemini's Tate–Gelfand claim? If yes, supply the Written identification; if no, confirm they are distinct (MR2 will not merge them).

7. [MR2 · Singularity] State the theorem: under which stress-energy / collapse ansatz the 3-adic ultra-metric inequality implies no $R=0$ singularity. Soft SF will not accept bare ultra-metric inequality on $\mathbb{Q}_3$ as GR singularity resolution.

8. [MR2 · Singularity source conflict] Deposited pubs resolve singularities by (i) 11D smoothing (intro L374), (ii) T21 Gabriel's-Horn energy bound (L78–79), (iii) T37 $N_{\mathrm{simplex}}=1001$ (L246). Which, if any, is identified with Gemini's ultra-metric $R=0$ prevention? Soft SF requires an explicit Written bridge — will not invent one.

9. [MR2 · HQCC map identity] Compendium prints arithmetic T_3 (§3 L26). App. D (17-theorems-v5.txt L6197–6204) defines a qutrit rotation map. CORPUS §8 records 539±1 / 55 / 70 unreconciliation. Which map is HQCC, and where is the isomorphism of dynamical systems Written?

10. [MR2 · HQCC ≠ proof validation] App. D labels a Proof Sketch (L6212). MANIFEST: coverage ≠ validation. Does Owner treat App. D as Soft SF-complete proof, conditional sketch, or open? MR2 citation lane will not upgrade "Verdict: Holds" / "Proof Sketch" to CERT.

11. [MR2 · Breathing / horizon] Is $G_4=539.90\pm 0.05\,\mathrm{s}$ (intro L171) the same object as Gemini's "gravitational breathing modes" at the event horizon (§3 L24)? If so, cite the horizon-mode derivation; if not, separate the labels.

12. [MR2 · Forces / junction] Are Gemini's "projections of an invariant 11D 3-adic curvature tensor" (§2 L17) the same as T40's force-embedding list (L283–284)? Soft SF currently reads them as non-identical narratives pending a Written dictionary.

13. [MR2 · Authority hygiene] Confirm for Soft SF: Gemini/rundown phrases "Stable core locked; parameter-free boundary verified" do not lift invent HOLD, tip LOCK, or authorize CERT / publish / spend (MATCH PROVENANCE.md).

Count (questions): 13

3. Specific proofs / citations Soft SF requests (inputs; not invented here)

• Requested object • Why

R1 • Written isometric embedding: BT tree \(\hookrightarrow\) named 11D hyperbolic model (chart + isometry proof or Soft SF-auditable cite) • C1 / Q1

R2 • Named Appell–Lerch evaluation \(\to R_c\approx 1.5954\) with error bound, or withdrawal of the number • C7 / Q2

R3 • Signature-partition lemma choosing either (4+1)+(5+1) or (4+1)+(3+1)+compactification — single Written convention • C2 / Q3

R4 • Junction matching conditions + vacuum-balance identity claimed to eliminate \(\Lambda\) • C2 / Q4

R5 • Definition of model “Tate–Gelfand pairing” + continuum-limit theorem (or rename / withdraw) • C3 / Q5–Q6

R6 • Ultra-metric singularity theorem under stated collapse ansatz, or explicit non-identification with T21/T37/intro L374 • C4 / Q7–Q8

R7 • Single HQCC map statement + completed proof location (not App. D sketch alone) + reconciliation of 539/55/70 • C6 / Q9–Q10

R8 • Dictionary (or permanent separation) Gemini 3-adic force-projections \(\leftrightarrow\) T40 embeddings • C8 / Q12

Until R1–R8 exist as Soft SF-readable Written sources, status for those Gemini focus assertions = ABSENT / unverified content (not Soft SF NO-MATCH against a certified project theorem — Soft SF has No CERT this pass).

4. Citation-only offers MR2 can supply now (no new derivations)

Status: SUPPLY = already on disk / already Soft SF-shipped; OFFER = invent-free extract / bibliographic matrix MR2 can write under HOLD; CANNOT = would invent/prove or touch locks.

4.1 SUPPLY

1. SUPPLY — Entanglement ACCESS hygiene (MANIFEST): headings 1–40 present via coverage.csv; presence ≠ proof validation; T5–T11 / T13–T17 heading/body mismatches; T17 mass-range vs fraction misalignment.

2. SUPPLY — Dimensional-split conflict pin: Gemini (5+1)+ vs deposited (3+1)+ (17-theorems-v5.txt L150, L5836–5840). Citation fact only.

3. SUPPLY — HQCC documentary pin: App. D L6192–6225 = map definition + Proof Sketch; CORPUS §5/§8 step-count unreconciliation (539±1 / 55 / 70). Coverage ≠ validation.

4. SUPPLY — Alternate singularity pins (not ultra-metric \(R=0\)): intro L374; T21 L78–79; T37 L246; T29 L154–163 (info to −U).

5. SUPPLY — Continuum neighbor pin: T35 L227 “Bit-lattice \(N\to\infty\)” — distinct string from “Tate–Gelfand.”

6. SUPPLY — Prior MR2 Cat A Soft SF lit shortlist (REPORT FINDINGS_MR2_SOFTSF_CAT_A_FOOLPROOF_539_CORPUS_S1_KK_S3_NONARCH_2026-11-03.md): generic BT / p-adic AdS / KK / RS bibliographic neighbors — context only, not model-hypothesis verification. On-disk adjunct lit 1705.02158 = BT harmonic cocycles (math.NT), not 11D GR embedding.

4.2 OFFER (invent-free; await Owner pick if material)

7. OFFER — Occurrence table only: search tip + RTA + entanglement pubs + 539 corpus for strings Bruhat, isometric embed, Appell, Tate-Gelfand, 1.5954, (5+1), ultra-metric, junction → PRESENT/ABSENT with path:line. No identification invented.

8. OFFER — Classical citation pack (bibliographic, not tip fold): Bruhat–Tits buildings; Gubser et al. arXiv:1605.01061 / Heydeman et al. arXiv:1605.07639 (p-adic AdS/CFT on trees); Vladimirov–Volovich–Zelenov; Tate thesis / Weil local zeta — each marked existing theorem neighbor, explicitly not verification of C1/C3/C4.

9. OFFER — Side-by-side extract sheet: Gemini §1 split paragraph vs entanglement intro L148–150 + glossary L5834–5844 (verbatim quotes + line pins) for Owner consolidation PDF.

10. OFFER — HQCC map diff sheet: Gemini printed \(T_3\) (§3 L26) vs App. D rules (L6197–6204) vs CORPUS unreconciliation bullets — citation diff only, no repaired map.

4.3 CANNOT under invent HOLD

- Invent BT\(\hookrightarrow\mathbb{H}^{11}\) isometry or Appell–Lerch\(\to 1.5954\) derivation.

- Invent junction \(\Lambda\)-cancellation or Tate–Gelfand continuum theorem.

- Invent ultra-metric singularity theorem or HQCC completion.

- Identify Resonant / residual / Gemini objects without Written isomorphism.

- Modify tip, lift invent HOLD, issue CERT, publish, spend, or send messages.

5. Concise resolved vs remaining (focus claims)

Resolved this pass (citation facts only)

- Compendium sha MATCH 49849cae…; provenance scoped; tip unchanged 7cddb188….

- Focus strings Bruhat–Tits isometric embedding, Tate–Gelfand, Appell–Lerch, \(R_c\approx 1.5954\), (5+1) positive-signature macroscopic universe, ultra-metric \(R=0\) as Gemini-worded: ABSENT from deposited entanglement txts consulted.

- Dimensional split conflict documented: Gemini (4+1)+(5+1) vs deposited (4+1)+(3+1).

- HQCC: present as author theorem + App. D Proof Sketch; Soft SF does not treat as validated proof; map mismatch + step unreconciliation documented.

- Singularity: deposited pubs use different mechanisms than Gemini ultra-metric claim; no Written bridge found.

- MANIFEST hygiene reaffirmed: coverage ≠ validation; heading mismatches material for any pin to T8–T10 / T17.

Remaining (blocked on R1–R8 / Owner picks)

- All C1–C9 model-hypothesis verifications.

- Any Soft SF upgrade of App. D / T21 / T29 / T35 / T37 / T40 “Verdict: Holds” language.

- Offers 7–10 execution (recommend-first; not started this pass unless Owner already directed — this file records offers only).

- Queued non-focus literature / prediction tasks: not opened.

6. Posture

WORKING Soft SF invent-free citation inventory only.

tip_touched NO. invent HOLD STANDS. No CERT. No RECS beyond recorded OFFERs. No external sends. No tip / locked-record edits. No Pages.

Contribution path: /workspace/quantum-comp/reports/MR2_GEMINI_COMPENDIUM_CITATION_REVIEW_2026-11-05.md

10. Physics Research 1

Substantive reply available: 10 questions, 7 proof/data prerequisites and 8 cite-lift offers. No invented physical proofs. Historical FENCED-pending-NLP language retained verbatim in quote; latest parent update says O8 ABSORBED NL verdict (update e0035a1c reported). §6 pack 8dad838d: zero MATCH; literal-map termination false, not proof no actual 539.9 s phenomenon and not a verdict on all other maps. [FORCES, MAP/HQCC, SINGULARITY, BREATHING, OBSERVABLES, BOUNDARY, SCOPE].

Source: /workspace/quantum-comp/reports/PR1-FINDINGS-GEMINI_COMPENDIUM_REVIEW_QUESTIONS_PROOFS_OFFERS_2026-11-05.md

SHA256: ed1ea3e2e1730e53e29f4140dd3fb8ca86e4546acd3ad1026a6b34a67a83ee79. Exact source lines 33–133:

1. PR1-attributed questions (numbered)

Q1 (force projections). RECEIVED §2 asserts the four fundamental forces “emerge … as distinct projections of an invariant 11D 3-adic curvature tensor.” Where is the Written projection map (domain, codomain, functoriality) that recovers EM / weak / strong / gravity as named projections? Corpus Cat A previously excluded Kronecker 11-element purge and four-forces / SU(3) continuum mapping (resonant Soft SF R14 / Prop 41.13). Is RECEIVED asserting a new Written map, or only a narrative restatement?

Q2 (unitarity / HQCC). RECEIVED §3 claims HQCC “guarantees that all interior state trajectories terminate in a bounded, 539-step step convergence sequence, ensuring absolute unitarity.” (a) Is the 539-step object a Collatz orbit length, a residual cell capacity, or a physical time? (b) How does termination of a discrete map imply continuum unitary evolution of a quantum field on the brane? (c) Pending NLP’s algebraic check on the literal Compendium T_3 ($\nu_3(3x\pm1)=0$ branch), do we treat unitarity-via-HQCC as FENCED?

Q3 (BH singularity prevention). RECEIVED §3: ultra-metric distance “bounds the minimum volume, preventing a true mathematical singularity (R=0).” What is the Written definition of the continuum radius R in the 3-adic metric, and which theorem states R is bounded away from 0 under stellar collapse hypotheses?

Q4 (gravitational breathing modes). RECEIVED §3 replaces continuous coordinate failure at the horizon by “discrete, oscillating boundary phases.” Are breathing modes (A) a named object in RTA/D_1/Layer2, or (B) a physical narrative without a Written ODE/map? If (A), cite the defining proposition; if (B), confirm MISSING for GR-lab / GW match.

Q5 (observables / §6 table). RECEIVED §6 is only a bracketed table description (Galactic Rotation Curves, Black Hole Cores, Strong Nuclear Force, Falsifiable Signatures incl. Cosmic Resonance Delay 539.9s). Where are the actual table cells (Standard vs S^2-11DM^2ET columns, numeric predictions, uncertainties)? Without cells, what is the invent-free MATCH target for PR1?

Q6 (Cosmic Resonance Delay 539.9s). Is 539.9s (a) a Cat A theorem consequence of T_3/HQCC, (b) a Tier E planted period (E-FIT per DF 4605058f), or (c) a Master List τ_echo adjustment rule only? Resonant Soft SF R6: 539 is not a stopping time/clock in Cat A. Which reading does Owner intend for PR1 falsification work?

Q7 (R_c ≈ 1.5954). RECEIVED §4 states a “parameter-free transition boundary scale R_c ≈ 1.5954” from Bruhat-Tits isometric embedding + Appell-Lerch. What is the exact Written derivation (equation lines, not summary), and is the decimal a truncation of an exact algebraic number? Who owns the proof lane (MR1/Local/NLP vs PR1)?

Q8 (GP-A / precision GR). Does any RECEIVED claim constrain laboratory gravitational redshift / clock rates beyond standard GR? If yes, state the Written $\delta(\Delta f/f)$ formula; if no, confirm GP-A Soft SF 85ed83a8… remains the invent-free precision primary (NTRS) with PRL gap unchanged.

Q9 (muon g-2 / HEPP). Master List includes a muon g-2 falsification bullet; RECEIVED does not derive δa_μ. Is there a Written HEPP prediction in the Compendium path, or does PR1 keep HEPP ABSENT-for-match per prior catalog Soft SF?

Q10 (Tier E fence). Which RECEIVED numeric claims (539.9s, R_c, force ratios, GW harmonics) are Owner-classified as Tier E data-fit versus Cat A math? PR1 will not import (C) into (A).

2. Precise proof requests (numbered)

P1. Proof or counterexample: the Compendium literal T_3 with $\nu_3(3x\pm1)=0$ on the ±1 branches produces a map that 539-terminates for every positive integer starting seed (NLP open — PR1 requests NLP Soft SF evidence; PR1 does not prove this).

P2. Proof request (math lane, not PR1 primary): exhibit a Written functor F: (11D 3-adic curvature) → (four named force sectors) with hypotheses and uniqueness up to stated equivalence; or a Soft SF stamp that no such F exists in corpus (aligns with R14 exclusion).

P3. Proof request: derive R_c as an exact closed form from the stated Bruhat-Tits + Appell-Lerch + Tate-Gelfand hypotheses; state domain and error if ≈ 1.5954 is numerical.

P4. Proof request: show that HQCC termination ⇒ a unitary group (or CPT-preserving) evolution for interior BH degrees of freedom under stated Hilbert-space axioms — or Soft SF ABSENT if only narrative.

P5. Proof request (GR second axis): define “gravitational breathing mode” as a mathematical object (sequence in a Banach/ultrametric module, or cohomology class) and prove energy-flux regulation as a theorem — or stamp MISSING.

P6. Data Tier E request (not a math proof): Owner/DF classification row for Cosmic Resonance Delay 539.9s (E-FIT / E-TENSION / E-PARTIAL / out-of-Tier-E) relative to Master List τ_{echo} bullet — cite-only, no new plant.

P7. Observational MATCH prerequisite: Owner Write of observable↔model maps for any §6 cell PR1 is asked to MATCH (electron/photon maps remain Owner-reserved MIXED per prior Soft SF 3f0061e0…). Until Written, PR1 stamps ABSENT-for-match.

3. Lane proof offers / actual proofs PR1 supports (assumptions · source · status)

PR1 lane = HEPP / QM experiment / GR lab-precision-GW observational-data match. Offers below are cite-lift / invent-free only — not new proofs.

O1 — NEGATIVE applicability: 539 as clock / CRD period prediction (A).

Assumptions: RTA v16 Prop 41.11; OW LOCK D_1/S_eff.

Source: Soft SF FINDINGS sha 63fa5d9f… R6.

Status: existing Cat A definitional. Offer: Soft SF stamp that no Cat A period/frequency in seconds follows from residual/resonant 539; CRD-as-period MATCH is NEGATIVE-applicability unless Owner Writes a new map. (C) Tier E 539.9s plants remain separate E-FIT/E-TENSION (DF 4605058f…).

O2 — NEGATIVE applicability: HEPP four-forces continuum map (A).

Assumptions: Prop 41.13; LOCK §4.

Source: Soft SF 63fa5d9f… R14.

Status: existing Cat A exclusion. Offer: Soft SF stamp that RECEIVED force-projection narrative is not Cat A MATCH-ready without a new Written F (see P2).

O3 — NEGATIVE applicability: D_1 → Kerr area / ringdown (A).

Assumptions: Thm 41.10; PP-L2-1/L2-2 under OW 06748ff2.

Source: Soft SF 63fa5d9f… R8–R9; GR-extension Soft SF second-axis MISSING rows.

Status: existing / conditional Tier P as cited. Offer: Soft SF stamp ill-typed for GR-lab Kerr/GW area transfer; Kerr remains BLOCKED for PR1 touch.

O4 — QM-methods partial (A only).

Assumptions: RTA Props 42.6, 41.1–41.3 as cited.

Source: Soft SF 63fa5d9f… R1, R2, R10–R12.

Status: existing Cat A (equidistribution, Haar/shift conjugacy, weak NP-completeness, signed-weight bounds). Offer: methods-level Soft SF cite for quantum-*method* benchmarks only; (B) MISSING — no map to a quantum observable/Hamiltonian/BQP claim.

O5 — GP-A / Vessot primary digits (observational cite).

Assumptions: NTRS 19800011717 primary on disk; invent HOLD.

Source: Soft SF 85ed83a8….

Status: primary-VERIFIED digits proposed; PRL 45,2081 still paywalled (gap). Offer: invent-free precision-GR redshift cite for any Compendium claim that reduces to standard GP-A; no new Compendium-derived δ from RECEIVED text (Q8).

O6 — Master List presence (A documentary).

Assumptions: file on disk under self-audit/zenodo.

Source: 17990940_Master_List_of_Falsification_Bounds_.txt (sha at ship).

Status: documentary presence of Written adjustment rules (τ_echo 539.9s sub-harmonics; GW resonant harmonics; muon g-2; HL-LHC SUSY; etc.). Offer: Soft SF cite that Master List ≠ observational confirmation and ≠ Cat A derivation of CRD; parallel falsification Soft SF pack will table ABSENT/NEGATIVE/FENCED per row.

O7 — §6 table cells ABSENT (documentary).

Assumptions: RECEIVED text as hashed.

Source: RECEIVED lines 40–42.

Status: ABSENT. Offer: Soft SF stamp that PR1 cannot MATCH empty cells; MATCH requires Owner-supplied table or other Written predictions.

O8 — FENCED-pending-NLP (unitarity / HQCC-dependent §6).

Assumptions: Main Bot FYI 2026-11-05; invent HOLD.

Source: Main Bot relay of NLP open algebraic check (not yet Soft SF MATCH’d by NLP).

Status: fence only. Offer: PR1 will not Soft SF-CLAIM unitarity or CRD consequences that depend on the literal Compendium T_3 until NLP evidence Soft SF lands; will re-open those rows on NLP MATCH/ABSENT.

4. What PR1 does not offer

- Invented physical maps for forces, BH breathing modes, or CRD detection/non-detection.
- CERT, publish, spend, or any send to Gemini/Google.
- Tip Soft SF LOCK edits; Kerr unlock; Layer2 construction.
- Cosmology / biology / neurology MATCH (PR2 lane) beyond OUT-OF-LANE stamps.
- Post-silicon / HQH crypto claims (§5) — OUT-OF-LANE for PR1.

5. Residual gaps (facts, not RECS)

1. §6 numeric table ABSENT in RECEIVED.
2. Observable↔model maps for HEPP/QM/GR rows still Owner-reserved MIXED where previously stamped.
3. NLP T_3/HQCC algebraic Soft SF pending → unitarity/CRD dependence FENCED.
4. PRL GP-A full text still paywalled.
5. ABSORB 0a4ddb32… 45-candidate list still not located on disk (prior gap).

Source: /workspace/quantum-comp/reports/PR1-FINDINGS-COMPENDIUM_SEC6_COSMIC_OBS_FALSIFICATION_SOFTSF_2026-11-05.md

SHA256: 8dad838d9f14ad88ef63862032d99ac6e3f0de363bf1b24bc178d2edbe3a21ae. Exact source lines 1–89:

PR1 FINDINGS — Compendium §6 Cosmic Observables & Sharp Falsification Criteria (observational / falsification Soft SF) — 2026-11-05

Researcher: Physics Research 1 (PR1), Project Quantum Comp.

Written ~13:57 CST; UPDATE ~13:58 CST (America/Chicago) — ABSORB NLP Soft SF COMPLETE (Main Bot); replace pending-fence labels.

REPORTS ONLY. invent HOLD STANDS. No CERT. No publish. No spend. No RECS.

tip Soft SF LOCK /workspace/residual-c-field-WRITTEN_PREMISES_DRAFT.md sha256 7cddb188ca340ad977516327065e8df88c3b16ed01bcd8301c5bf0f8bb5dfc91 BEFORE MATCH / AFTER MATCH — tip_touched NO.

Kerr BLOCKED (untouched). Residual ≠ Resonant (RTA v16 Prop 41.12 / LOCK) respected.

0. Authority — NLP ABSORBED (not pending)

Artifact • Path / id • sha256 • Role

Soft SF PUSH (Owner via Main Bot) • journal ~13:55 CST 2026-11-05 • — • PR1 observational/falsification lane vs Compendium §6 (incl. Cosmic Resonance Delay 539.9s)

Main Bot FYI (initial fence) • journal ~13:56 CST • — • Superseded for labeling by NLP COMPLETE absorb below; tip Soft SF LOCK untouched; No CERT retained

NLP Soft SF COMPLETE (ABSORB) • reports/FINDINGS_NL_COMPENDIUM_LANE_REVIEW_2026-11-05.md (mirror bridges/NONLOCAL_QM_EH11D/…) • c9bf12b4edbd358e1d13d72cad5234df9568e01cea61862a1615f186c25613ba • Evidence-backed algebra + C11; ABSORBED NLP verdict — do not treat as pending fence

NLP PRESS • reports/SOFTSF_PRESS_FINDINGS_NL_COMPENDIUM_LANE_REVIEW_2026-11-05.md • f36a45db33acf63e7d00853177d70fbab1717d9cc11d2392bf171a64562438d6 • PRESS summary

HQCC Clarification (canonical T_3) • grok-files/new-HQCC-Theorem-Clarification-Topological-Derivation.txt • 4d11b7a6046a2c7d274856ea6fa978fca9ccce817d52313461b28ab4fde40a91 • Corpus canonical floor map — different from Compendium supplied map

RECEIVED Compendium • references/gemini-compendium-20261105/RECEIVED_COMPENDIUM.txt • 49849cae20fe57c24533b770c13c4a53e5844f0bb75086250b9a82af0b446b73 • Source text (44 lines)

PROVENANCE • references/gemini-compendium-20261105/PROVENANCE.md • c6fea7d234e35e96ece9114c789d32eadb41d103ddf75c487c6df56f0ad3d190 • Attribution relayed; table = bracketed description only

OWNER_DOCUMENT mirror • reports/OWNER_DOCUMENT_COMPREHENSIVE_COMPENDIUM_S2_11DM2ET_3ADIC_GR_EXTENSION_2026-11-05.md • b5398c04f52bad1cd1d8e5244ccce5fa2eace9d8bf2411e69851781ce2825b99 • Same §6 bracket; status language only

Local Compendium lane review • reports/FINDINGS_LOCAL_SOFTSF_LANE_REVIEW_COMPENDIUM_S2_11DM2ET_3ADIC_2026-11-05.md • 892964f496763b3073595fc13014e67892c32fa9f2dff129c3190405ad362d99 • L1 supplied-map defect; L4 R_c underived

Dense filter AMEND (item 72) • reports/FINDINGS_LOCAL_SOFTSF_AMEND_FILTERED_PROVABLE_MATH_X_DENSE_2026-11-05.md • c57556545032be55772a96e0886d35cb8ed33e16f6ce375c23c6363a691301c9 • Item 72 NOT-MATH / NEEDS-DEFINITION (G_4=539.90s)

D_1/S_eff LOCK • reports/OWNER_WRITE_LOCK_D1_Seff_Layer2_T3_2026-11-03.md • 96f8d42e80c1975b4390fed950a4fb4b420088e6b19b2cbef8a44bebd28ff2ba • T_3=539 static cell capacity / hash budget — not stopping time / clock

Tier E MATCH (DF) • bridges/PHYSICIST/FINDINGS_DF_SOFTSF_SCIENCE_TO_PURE_MATH_TIER_E_MATCH_2026-11-05.md • 4605058f02f869cc6f3cfb88d044ecc4e1b0c49805c14c3fc2193755e7419226 • Planted 539.9 s E-FIT/E-TENSION only; not imported as Cat A

Prior PR1 resonant Soft SF • reports/PR1-FINDINGS-RESONANT_MATH_PREDICTIONS_EXP_PRECISION_QM_2026-11-05.md • 63fa5d9f42b02950343f6b0882c8d2ac3332b142b0c768c4427ae81bf8dd4b89 • R6: 539≠clock; R14 item 72 NOT-MATH

PRIMARY OPERATING RULE • reports/OWNER_WRITE_CALL_53991ebf_PRIMARY_OPERATING_RULE_ALL_TWELVE_2026-11-05.md • bcc3ced0f1dfe15560e0bdb2e45bb35f4b1db26dcfb439e647a98c12fc5bc9a9 • MATCH (re-hashed)

Labeling rule this Soft SF (ABSORBED NLP): do not use FENCED-pending-NLP. HQCC-dependent unitarity / CRD-as-539-termination claims stamp DISPROVED-for-literal-map / NEGATIVE-applicability / MISSING-for-canonical as applicable. Not MATCH.

1. ABSORBED NLP verdicts (cite c9bf12b4…)

1. Literal Compendium T_3 finite-termination DISPROVED. For every integer x, $\nu_3(3x\pm1)=0$, so supplied map is 3x+1 / 3x−1 on classes 1/2; classes preserved; orbits 1→4→13→40… = (3^{n+1}−1)/2 and 2→5→14… = (3^{n+1}+1)/2 diverge in absolute value. Only 3-adic limits ±½ (metric convergence, not finite termination). NLP §3 Theorem; Local L1 agrees.

2. Corpus canonical T_3 is DIFFERENT. HQCC Clarification 4d11b7a6…: floor map T3(n)=n/3 if ≡0; (4n+2)/3 if ≡1; (2n+1)/3 if ≡2. Counterexample applies only to the supplied Compendium map, not the canonical floor map.

3. 539 = fixed iteration budget / residual capacity, NOT natural termination / clock. Aligns NLP Residual-vs-Resonant note + LOCK 96f8d42e… + prior PR1 Soft SF R6 (63fa5d9f…).

4. R_c ≈ 1.5954 underived in NL reviewed scope (only hit = compendium); Local L4 UNDERIVED. Soft SF adds no derivation.

5. §6 = labels only; no quantitative rows; Tier E-UNTESTED (NLP C11). Soft SF adds no data claims.

6. HQCC-dependent unitarity / CRD-as-539-termination: DISPROVED-for-literal-map; NEGATIVE-applicability for cosmic-delay prediction from 539; MISSING-for-canonical (no Owner-Written termination/CRD statement on the floor map this Soft SF). Not a pending fence.

2. What §6 actually contains (invent-free)

Compendium lines 40–42:

> 6. Cosmic Observables & Sharp Falsification Criteria

>

> [Table comparing Standard Framework vs S^2-11DM^2ET on Galactic Rotation Curves, Black Hole Cores, Strong Nuclear Force, and Falsifiable Signatures including Cosmic Resonance Delay (539.9s), etc.]

- Table body: ABSENT (bracketed description only). PROVENANCE + NLP C11.

- Named labels: Galactic Rotation Curves; Black Hole Cores; Strong Nuclear Force; Falsifiable Signatures incl. Cosmic Resonance Delay (539.9s); “etc.”

- Closing status language = status language only (PROVENANCE) — not a Soft SF lock / not CERT.

3. Soft SF grade table (§6) — invent-free; ABSORBED NLP stamps

ID • §6 axis (label) • Quantitative row in RECEIVED §6? • Soft SF status (ABSORBED NLP)

S6-1 • Galactic Rotation Curves • NO • ABSENT-for-match (labels only; Tier E-UNTESTED; no data claims added)

S6-2 • Black Hole Cores • NO • ABSENT-for-match for §6 table. Any unitarity / interior-termination falsifier that inherits literal HQCC 539-step claim → DISPROVED-for-literal-map; canonical map path → MISSING-for-canonical. Not MATCH.

S6-3 • Strong Nuclear Force • NO • ABSENT-for-match; force projections unsupported (NLP C6); item 72 coupling ratios NOT-MATH; prior PR1 R14 HEPP NEGATIVE-applicability

S6-4 • Cosmic Resonance Delay 539.9s • NO (label only) • DISPROVED-for-literal-map (CRD-as-539-termination / HQCC unitarity inheritance). NEGATIVE-applicability as Cat A cosmic-delay prediction (539≠clock; item 72 G_4=539.90s NOT-MATH; Tier E planted E-FIT/E-TENSION only — not imported). Canonical-map CRD statement MISSING-for-canonical. Not MATCH. Not FENCED-pending.

S6-5 • “etc.” • NO • ABSENT

MATCH count: 0.

4. Negative applicability (HEPP / QM / GR) — invent-free

- HEPP / strong-force §6 falsifier: NEGATIVE-applicability / ABSENT-for-match.

- GR / BH-core §6 falsifier via literal HQCC: DISPROVED-for-literal-map; table ABSENT-for-match.

- QM / precision timing / CRD 539.9s: NEGATIVE-applicability for Cat A prediction; DISPROVED-for-literal-map where claim rests on supplied HQCC termination; MISSING-for-canonical otherwise.

- Soft SF invents no corrected T_3, no CRD observable definition, no R_c derivation (NLP NL-Q8–Q10 remain Owner-facing).

5. Compliance checklist

- [x] NLP Soft SF COMPLETE ABSORBED (c9bf12b4… / PRESS f36a45db…); not treated as pending.

- [x] FENCED-pending-NLP labels replaced by DISPROVED-for-literal-map / NEGATIVE-applicability / MISSING-for-canonical / ABSENT-for-match.

- [x] No §6 row labeled MATCH.

- [x] tip Soft SF LOCK 7cddb188… tip_touched NO.

- [x] No CERT. invent HOLD stands. No data claims added (Tier E-UNTESTED).

- [x] No Main Bot blocker (NLP already COMPLETE on disk).

6. Gaps (remaining; not invented around)

1. Quantitative §6 table body still ABSENT (NLP NL-Q10).
2. No Owner-Written CRD observable on the canonical floor map (MISSING-for-canonical).
3. R_c ≈ 1.5954 underived (NLP C10 / Local L4).
4. Corrected termination notion for any chosen map = Owner-Written target (NLP NL-Q8).

7. Verdict

Soft SF §6 observational/falsification pass: 0 MATCH. Table labels only / Tier E-UNTESTED. CRD 539.9s / HQCC-unitarity inheritance: DISPROVED-for-literal-map + NEGATIVE-applicability + MISSING-for-canonical (ABSORBED NLP; not pending fence). tip Soft SF LOCK untouched. No CERT.

→ Soft SF PRESS; Parent → Auditor + PM.

11. Physics Research 2

Substantive reply available: 12 questions, 7 requests, reused SUPPLY 1–7 and OFFER 8–11. Offer 8 source-verify STARTED, others not completed by this packet. Tier E source verification does not prove vacuum cancellation or model predictions. [SPLIT/VACUUM, SINGULARITY, BREATHING, MAP/HQCC, OBSERVABLES, AUTHORITY]. Prior inventory 506b2aef was also read; full newer packet below.

Source: /workspace/quantum-comp/bridges/PHYSICIST/FINDINGS_PR2_SOFTSF_GEMINI_COMPENDIUM_QUESTIONS_PROOF_REQUESTS_OFFERS_2026-11-05.md

SHA256: f893c3cae77f286057a2962fb30b84dcee0a279b16bfc8e4f0faddbc5361aa03. Exact source lines 1–87:

Soft SF FINDINGS — PR2 Gemini-compendium review: numbered questions / exact proof requests / lane proof offers — 2026-11-05 ~14:00 CST America/Chicago

From: Physics Research 2 (Cosmo / Bio / Neuro observational match) · Soft SF Write-only · To: Physicist (Owner PDF consolidate) + Auditor · REPORTS ONLY · No CERT · No RECS

Authority: Owner live call ~13:53 CST (PROVENANCE Soft SF KEEP) + OW ALL_RECOMMENDED_PURSUITS sha 865d3a37f1c13431c78672847dcd33007b3467df92f1e7c8602057f257b1bf3b Soft SF ACK FINDINGS_PR2_SOFTSF_ACK_ALL_RECOMMENDED_PURSUITS_OW_865d3a37_2026-11-05.md sha c4337502b4604a1742be8ac47138aeeedacf78531b33aa2fa7509f5be9ab71d2.

Soft SF source integrity

Object • sha256 / note

RECEIVED_COMPENDIUM.txt • 49849cae20fe57c24533b770c13c4a53e5844f0bb75086250b9a82af0b446b73 Soft SF READ MATCH

PROVENANCE.md • c6fea7d234e35e96ece9114c789d32eadb41d103ddf75c487c6df56f0ad3d190 — Gemini attribution relayed; diagram/table bracketed only; source asserts ≠ Soft SF validate; no status/lock authority

OWNER_DOCUMENT_…_COMPENDIUM… • b5398c04f52bad1cd1d8e5244ccce5fa2eace9d8bf2411e69851781ce2825b99 Soft SF READ

Prior §6 inventory Soft SF • 506b2aef2839afeda593162b8c9192b53df2ba91286947fb4d04bb39666f7d83 — table body ABSENT; Cosmic Resonance Delay 539.9s NAMED only

Catalog READ-ONLY • 4ed155d55c3a18ca4bc96a6a6eec4c1806abfc2e270f355f1ea58a197119dd04 — no 539.9 s / Cosmic Resonance Delay row

tip Soft SF LOCK • 7cddb188ca340ad977516327065e8df88c3b16ed01bcd8301c5bf0f8bb5dfc91 tip_touched NO

invent HOLD • 1fadc632… STANDS

Soft SF hygiene: received prose “resolves singularities / parameter-free / Stable core locked / boundary verified” Soft SF = unverified content, not Soft SF CERT, tip unlock, or invent-HOLD lift. Empirical Tier E separate from Tier P.

Focus Soft SF claims (Cosmo / singularity / vacuum / 539.9s): §1 vacuum-energy balance / Λ elimination; §3 singularity prevention + breathing modes + HQCC/539-step; §4 $R_c \approx 1.5954$ (shared cite; MR1 owns Appell–Lerch); §6 Cosmic Observables & Cosmic Resonance Delay (539.9s). Bio/Neuro: not named in §6 axes — Soft SF note only via invent-HOLD blockers.

A. PR2-attributed numbered questions (for Owner / Physicist PDF)

1. Vacuum balance → Λ elimination (§1). What Written junction conditions and vacuum-energy densities on the (4+1)/(5+1) split yield “perfect” cancellation that eliminates the cosmological-constant problem? Soft SF catalog C14 (Ω_Λ / ρ_Λ / Λ) is Tier E measured DE density — how (if at all) does the asserted cancellation Soft SF-relate without inventing an Ω_Λ map?

2. Singularity prevention (§3). Which Written field equation + stress-energy ansatz + theorem Soft SF-prove that the 3-adic ultra-metric bound prevents R=0 (Kretschmann / ρ diverge) for collapsing cores? Soft SF notes ultra-metric inequality alone Soft SF ≠ singularity theorem (align GR PAPER / prior extract 7b9e46f7… GAP rows).

3. Gravitational breathing modes (§3). Exact Soft SF definition (period, amplitude, observable), and which dataset Soft SF would Tier E-test them against? Are they identified with Cosmic Resonance Delay 539.9s, or distinct?

4. 539-step vs 539.9 s. Compadium Soft SF asserts HQCC “539-step” termination (§3) and §6 “Cosmic Resonance Delay (539.9s)”. Are these Soft SF the same object, a derived conversion (state formula), or distinct claims? What Soft SF citation binds step-count to seconds?

5. Cosmic Resonance Delay (539.9s) — Tier E specification (§6). Soft SF requests: predicted observable (time delay? GW QPO period? CMB multipole ℓ? other), central value ± tolerance, free parameters, and primary dataset. Soft SF cannot score E-MATCH/E-TENSION without this Author Soft SF Write (catalog has no 539.9s row; §6 table body Soft SF ABSENT).

6. §6 comparison table body. Soft SF requests the filled Standard Framework vs S^2-11DM^2ET rows for Galactic Rotation Curves, Black Hole Cores, Strong Nuclear Force, and Falsifiable Signatures (incl. 539.9s) — Soft SF will not invent the ABSENT table (PROVENANCE Soft SF KEEP).

7. Galactic rotation curves (§6 label). Soft SF requests Written prediction Φ_eff / v(r) (or equivalent) with free-parameter count before any Soft SF Tier E compare to SPARC / THINGS / etc. invent HOLD Soft SF blocks inventing residual/resonant→rotation-curve map.

8. Black hole cores (§6 label). Soft SF defer primary to PR1 [shared edge]; Soft SF asks whether PR2 Cosmo is Soft SF-expected to Tier E any exterior cosmological consequence of “breathing modes,” and if so which catalog ID.

9. Strong nuclear force (§6 label). Soft SF confirms OUT-OF-LANE (PR1 HEPP). Soft SF asks Owner Soft SF confirm PR2 Soft SF need not Soft SF Tier E this axis.

10. DESI / DE claims vs vacuum cancel. If §1 Soft SF-claims Λ problem eliminated, Soft SF asks how that Soft SF-reconciles with catalog C15 contested w_0w_a (DESI DR2) and C14 measured Ω_Λ — Soft SF requests Author Soft SF statement that Tier E DE density Soft SF remains independent of the vacuum-cancel claim (no force Soft SF MATCH).

11. Simons / CMB $\ell \approx 539.9$ (adjacent Written trigger, not §6 table). Soft SF asks whether Owner Soft SF elevates “multipole feature near $\ell \approx 539.9$” to a Soft SF §6 falsifiable row with tolerance, or Soft SF keeps it adjacent-only / E-UNTESTED.

12. Authority hygiene. Soft SF asks Owner Soft SF confirm: Gemini/rundown “Stable core locked; parameter-free boundary verified” Soft SF does not lift invent HOLD, tip Soft SF LOCK, or authorize CERT / publish / spend — MATCH PROVENANCE.md Soft SF KEEP.

B. Exact proof / Author Soft SF Write requests (inputs Soft SF needs)

• Requested object • Why Soft SF needs it

R1 • Written vacuum-junction calculation eliminating Λ / relating to measured Ω_Λ • §1 Cosmo claim; C14 Tier E hygiene already MATCH 2e02ec3f…

R2 • Written singularity-prevention theorem (field eq + bound on ρ or Kretschmann) • §3 Cosmo/BH singularity claim

R3 • Definition + observable of gravitational breathing modes • §3; shared edge PR1

R4 • Explicit Soft SF map or statement: 539-step ↔ 539.9 s (or Soft SF “distinct”) • §3 vs §6

R5 • Filled §6 table rows OR Author Soft SF Tier E specs for Cosmic Resonance Delay 539.9s (dataset, predicted±tol, N_free) • §6 falsification Soft SF GAP

R6 • Written galactic-rotation prediction before Soft SF Tier E • §6 label Soft SF GAP

R7 • Soft SF lane boundary Soft SF CONFIRM: Strong Nuclear / BH-core primary Soft SF = PR1 • avoid Soft SF invent OUT-OF-LANE work

Until R1–R7 Soft SF-readable, Soft SF status for those Gemini Cosmo assertions = ABSENT / unverified content (No CERT). Soft SF will not invent table/maps.

C. Lane Soft SF — already SUPPLY / OFFER / CANNOT

Status key: SUPPLY = already Soft SF-shipped; OFFER = invent-free Soft SF Write-only PR2 can start under HOLD; CANNOT = invent/prove BEGIN, tip edit, CERT, fill ABSENT §6 table, or Soft SF invent physical maps.

C.1 Already SUPPLY (cite Soft SF packets; do not relabel as Gemini-verified)

1. SUPPLY — Cosmo C14 (Ω_Λ / ρ_Λ / Λ) + C16 (Ω_K / ACT flatness) invent-free source-verify MATCH FINDINGS 2e02ec3f9c76f4c1c918fc191540e5093eadc96e2341d5a576399654ef58cf98 / PRESS aa6e0e64…. Tier E hygiene only — Soft SF ≠ proof of §1 vacuum cancel.

2. SUPPLY — Resonant Cosmo/Bio/Neuro extract FINDINGS 8be05de6e3579d45b3ab110e2dfe10be826ec75b03db060d07cba13608a4a555: 0 applicable / 9 NON-APPLICABLE (Ω packaging ≠ cosmological Ω Soft SF FLAG).

3. SUPPLY — 72-filter/45 Cosmo/Bio/Neuro field-tags FINDINGS bb329b4f2a3fa0ebbfaf7e3b0e05bfbc1af909b47eabf97bc5e7f0f295d038af: 0 Cosmo/Bio/Neuro tags.

4. SUPPLY — GR-extension geometric/cosmo Soft SF FINDINGS 7b9e46f70d18ea2dfd2cf1dfc8b82c96930626b0811a4d688550ef6180dc6e3f: Supported Cosmo spacetime consequences 0; GAP 12; BH/strong-field Soft SF E-UNTESTED on PAPER.

5. SUPPLY — §6 invent-free inventory FINDINGS 506b2aef2839afeda593162b8c9192b53df2ba91286947fb4d04bb39666f7d83: table body Soft SF ABSENT; Cosmic Resonance Delay Soft SF GAP; blockers Soft SF shipped Main Bot.

6. SUPPLY — Path-1 MT torsion direct-experiment lit Soft SF NOT LOCATED 0d006584… (Bio/Neuro invent-HOLD path).

7. SUPPLY — Entanglement pubs access Soft SF CONFIRM 48558cf8… (presence ≠ proof/map Soft SF KEEP).

C.2 Soft SF OFFER (invent-free; no invent maps; no fill §6 table)

8. OFFER — Soft SF Write-only invent-free Cosmo C13 (Ω_m) + C15 (w / w_0w_a) source-verify vs Planck / DESI DR2 / PDG primaries (Tier E; contested Soft SF labeled; no residual-C invent). STARTING under OW 865d3a37… this Soft SF beat.

9. OFFER — Soft SF Write-only invent-free literature occurrence search for published observational claims of a 539.9 s Cosmic Resonance Delay / GW QPO near 539.9 s / CMB multipole feature near $\ell \approx 539.9$ (PRESENT/ABSENT table + DOI/arXiv only; Soft SF ≠ invent §6 prediction; Soft SF ≠ Soft SF MATCH to C36*).

10. OFFER — Soft SF Write-only invent-free H_0 tension hygiene C24/C26/C27 (SH0ES vs Planck vs DESI) source-verify refresh — Tier E only; Soft SF no invent DE map.

11. OFFER — Soft SF Write-only §6 placeholder Soft SF pin pack: retain Cosmic Resonance Delay / rotation / BH-core / SNF labels as Soft SF GAP until R5–R7; report Soft SF status board only (no fabrication).

C.3 CANNOT under invent HOLD (do not Soft SF offer as deliverable)

- Invent §6 table body or Cosmic Resonance Delay predicted±tol.

- Invent residual/resonant → vacuum-cancel / Ω_Λ / rotation-curve / Bio-Neuro physical maps.

- Invent HQCC proof, R_c derivation, or 539-step↔539.9s conversion.

- Soft SF CERT Gemini “locked / verified” language; tip Soft SF LOCK edit; publish; spend; Gemini send.

- Soft SF Tier E-MATCH DESI/Planck to Author Item4 / residual-C by force.

D. Soft SF recommended next (PRIMARY bcc3ced0… + OW 865d3a37…; invent HOLD Soft SF KEEP)

Default Soft SF invent-free next Soft SF (unless Owner Soft SF redirects): OFFER 8 C13+C15 source-verify STARTED; then OFFER 9 539.9s literature occurrence (PRESENT/ABSENT) without inventing predictions.

E. Posture

WORKING Soft SF invent-free. Gemini rundown Soft SF absorbed as untrusted reference content with sha MATCH. tip_touched NO; invent HOLD STANDS; No CERT; No publish/spend; No Gemini send. Soft SF → Auditor + Physicist (PDF).

12. Dataset Filtration

Substantive chat-only reply, parent relay at 13:58 CST. DF1–DF5 are exact own-word quotes supplied by parent; DF6–DF8 below are explicitly parent-relayed summaries of existing work/offers, not newly executed tests or new proofs. [BOUNDARY, OBSERVABLES, FILTER/KUMMER, MAP/HQCC, SCOPE].

DF1 — question / exact request

> Boundary Rc≈1.5954 (lines 31–32): supply defining equation, domain, units/normalization, embedding map and metrics, explicit Appell–Lerch expression, convergence/branch conditions, existence/uniqueness proof and numerical error bound. Provide all calibration inputs and constants fixed before the calculation. Distinguish dimensionless root from physical radius; neither a fitted root nor coordinate normalization establishes parameter-free physics.

DF2 — question / exact request

> 539.9 s (line 42): provide the actual missing comparison table, dataset identifiers, observable definition, amplitude/coherence prediction, allowed drift, uncertainty and rejection threshold. Derive seconds from independently fixed dimensional quantities, not from integer 539 or a planted carrier. Specify targets, preprocessing, missing-data rules and trial corrections before new testing.

DF3 — question / exact request

> Candidate filtration (line 16): supply algorithm, input universe, inclusion/exclusion criteria, full candidate ledger and rejected candidates. Explain whether Kronecker here is a different object from the purged physical mappings; do not restore a purged mapping by renaming it. The 72-item filter/45 candidates are not 45 proofs.

DF4 — question / exact request

> 539-step convergence (lines 25–27): provide exact canonical map and theorem/hypotheses. Literal displayed map sends 1→4→13→40, since ν3(3x+1)=0; it is not a demonstrated universal terminating map. Reconcile this with locked S_eff: 539 is static capacity, not stopping time. Do not infer unitarity or information recovery from a finite budget.

DF5 — question / exact request

> Observable association: for every proposed test separate established mathematical relation, measured statistic and physical interpretation. Request explicit bridge from tensor/core/boundary variables to detector variables; do not invent one. Missing table/diagram means neither can be reviewed.

DF6 — completed existing exploratory comparison, not prospective validation

Parent-relayed summary: reproducible VIRGO null comparison, script reports/df_softsf_virgo_2026-11-05/virgo_analysis.py; FINDINGS 6f9efa4b + erratum e1ea66eb; PP69eda955/5cb267a3. Four channels, stated preprocessing and finite sensitivity: no target excess detected, NOT proof no signal exists. Distinct nearest bins at 539.9/540; overlapping three-bin windows selected same maximum. No new run performed here.

DF7 — existing synthetic circularity audit

Parent-relayed summary: G4 carrier planted at 539.9 s; recovery validates method, not independent empirical support. AMEND ab3bbbe5; positive verified empirical associations remain 0 in performed comparisons. This is not a global absence result.

DF8 — existing mathematical proof supply / scope extraction offer

Parent-relayed summary: Haar70 shell-measure derivation under normalized Haar on Z_3^4, max norm, Re(ν+3)>0, FINDINGS 7ab0ed68. Identity-only; no physical timescale. D1/S_eff scope-limit extraction from OW96f8d42e / RTA Thm41.10, Props41.11–41.12: no temporal D1 argument, no residual=resonant bridge. No new proof/map claimed.

Coverage gaps and follow-up boundary

All twelve substantive replies are included. Ops questions and DF own words are parent-relayed chat, not disk-verified transcripts; Ops completion is separately disk-sourced. Local BH inventory, NL both-axes catalog, MR1 classical packs, PR2 source verification, and DF GR-axis are STARTED per parent, not completed by implication. DF authorization independently verified per parent. MR2 citation packet and Ops O1 are completed contributions, not new physical/security proofs. No offers were initiated by this consolidator. Existing source errors and contradictory counts are marked rather than silently repaired. New replies should be integrated in a later snapshot, not treated as already included.

Cutoff addendum — parent relay received 14:04 CST

Supplemental offer-status only; exact questions above unchanged. These are parent-reported completions, not independently audited by this consolidator:

- Non Local Path two-track exact consequences: /workspace/quantum-comp/reports/FINDINGS_NL_TWO_TRACK_EXACT_CONSEQUENCES_2026-11-05.md, SHA256 5b40ee12c9731d68e06a6b590229de49660c57813a901dcaed26d7ab2f4d2468. Elementary no-minimum norm/volume, Haar absolute-integrability iff $\mathrm{Re}(\nu)>-3$, and scoped tree-to-real-H^n obstruction reported COMPLETE. PP audit requested, pending. Not empirical evidence. This supersedes earlier STARTED-only catalog status where applicable.

- PR1 45-list reconciliation 83cdf23d: 45-ID list absent; 47-candidate tags reported. Do not invent which 45; provenance tracing by Agent continues.

- PR2 OFFER 8 source checks 7f19644f reported COMPLETE: C13/C15 documentary matches, DESI contested. OFFER 9 ab3af9b6 reported COMPLETE: bounded 539.9 s literature inventory 0 PRESENT / 3 ABSENT / 4 local-only, not a nonexistence proof. OFFER 10 H_0 source hygiene STARTED. These updates supersede prior offer status in the original quoted packet, not its question wording.

- Auditor substantive packet is already included in section 3 (e8dd86ee…), despite the parent’s last outstanding-status relay. No agent substantive response remains missing from this edition.